

Introduction to clinical biomechanics

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Mo Farah has nine key elements to his running technique that have allowed him to become Britain's greatest ever runner.

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The term 'biomechanics' can be used in a variety of ways. In this book, biomechanics refers to the description, analysis and assessment of human movement during sporting activities.¹ Broadly, biomechanics can be broken into three categories, including 'kinematics' (movement we can see), 'kinetics' (forces driving the movement) and 'neuromotor' (muscle function controlling forces and movement). This chapter will focus primarily on the actual movement occurring in the body segments (kinematics). Our approach can be referred to as 'subjective biomechanical analysis'. We aim to describe movement such as running or squatting as it appears to visual observation. This reflects how clinicians assess and treat, and it can be done with the assistance of video analysis and without expensive laboratory equipment.

The aims of this chapter are to:

- outline the basics of 'ideal' lower limb biomechanics
- explain the ideal biomechanics with running
- describe lower limb biomechanical assessment in the clinical setting
- outline how to clinically assess footwear
- review the best available evidence associating biomechanical factors with injuries, as well as sharing clinical opinions as to which technical factors in sports contribute to specific injuries
- discuss how to manage biomechanical abnormalities detected in the assessment
- explain normal and abnormal upper limb biomechanics.

We address lower limb and upper limb biomechanics separately for the learner's convenience, but the experienced clinician will consider the close relationship between the upper and lower limbs during a variety of functional tasks.

Ultimately, biomechanical evaluation should be completed based on task specificity to ensure the clinician is confident in the accuracy of information obtained.

'IDEAL' LOWER LIMB BIOMECHANICS—THE BASICS

Here we discuss ideal structural characteristics, including available joint range of motion and stance position. Note that each individual has his or her own mechanical make-up due to structural characteristics (anatomy), and may never achieve the 'ideal' position or biomechanical function. Table 8.1 and Figure 8.1 provide reference for ideal joint range of motion and planes of movement respectively.

Lower limb joint motion

The *hip joint* is formed between the femoral head and acetabulum. The ball-and-socket structure of this joint permits motion in all three planes.

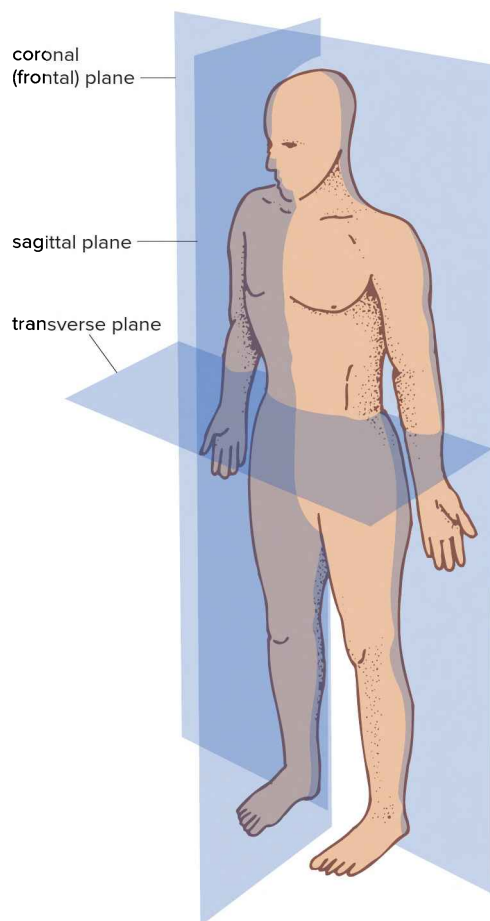
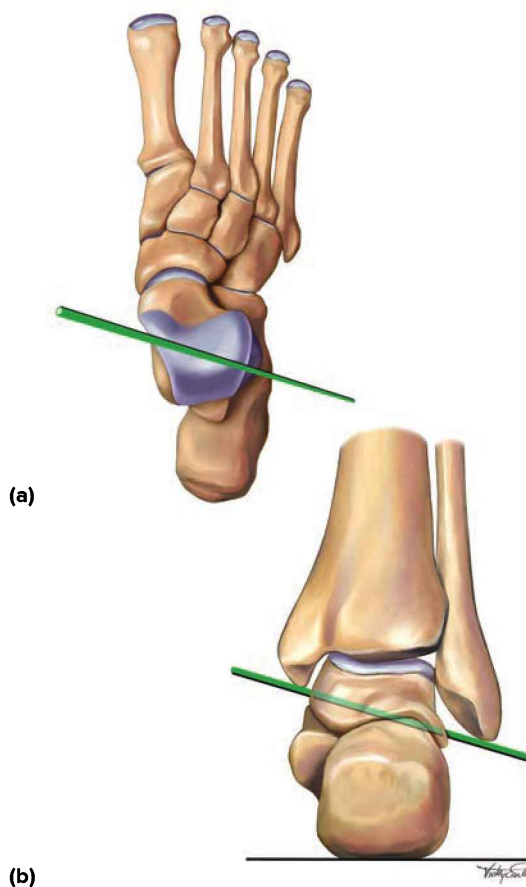
The *knee joint* is formed between the tibial plateau and femoral condyles. Primarily a hinge joint, the knee allows flexion and extension in the sagittal plane. The knee also permits some rotation in the transverse plane. This secondary motion is particularly important to allow the knee to lock into an extended position for stance stability, and to unlock when moving into flexion for shock absorption.

The *ankle joint* (between the shank and rearfoot) consists primarily of two articulations, the *talocrural joint* and the *subtalar joint* (STJ). The talocrural joint is formed between the talus and the mortise of the tibia and fibula malleoli. Its axis of motion is predominantly in the frontal plane which allows dorsiflexion and plantarflexion motion in the sagittal plane (Fig. 8.2a and b).²

Table 8.1 A guide to lower limb joint ranges of motion when in neutral positions

Joint	Plane	Assessment position	Available range
Hip	Sagittal	Supine	Flexion = 120°
		Prone	Extension = 20°
	Frontal	Supine	Abduction = 40°; adduction = 25°
	Transverse	Supine/prone±hip flexion/extension	Internal rotation = 45°; external rotation = 45°
Knee	Sagittal	Supine	Flexion = 135°; extension = 0°
	Frontal		None
	Transverse	Full extension	None
		70° flexion	45° rotation
Foot and ankle (tripplanar)*	Sagittal	Supine	Plantarflexion = 45°; dorsiflexion = 20°
	Frontal		Supination = 45–60°; pronation = 15–30°
First metatarsophalangeal (MTP)	Sagittal	Supine	Plantarflexion = 45°; dorsiflexion = 70°

*Refers to combined motion of the talocrual, subtalar, midtarsal and metatarsal break joints

**Figure 8.1** Anatomical planes of the body**Figure 8.2** Axis of motion of the ankle joint (a) Superior view (b) Posterior view

The STJ is formed between the calcaneus and talus. The three articular facets of the STJ allow for complex triplanar motions of pronation and supination. The axis of motion runs posteriorly and inferiorly in the sagittal plane ($40\text{--}50^\circ$), and laterally in the transverse plane ($20\text{--}25^\circ$)² (Fig. 8.3a-d). During pronation, the STJ axis provides primarily eversion, which is combined with dorsiflexion and abduction (Fig. 8.3d). During supination, the STJ axis provides primarily inversion, which is combined with plantarflexion and adduction (Fig. 8.3c).

The *midtarsal joint* is formed between the midfoot and rearfoot, and consists of two articulations, the *calcaneocuboid joint* and the *talonavicular joint*. These articulations provide two joint axes. The oblique axis allows large amounts of sagittal plane (dorsiflexion/plantarflexion) and transverse plane (abduction/adduction) motion, while the longitudinal axis allows small amounts of frontal plane (eversion and inversion) motion (Fig. 8.4a and b).

Importantly, the orientation between these two axes allows the role of the foot to change during

weight-bearing. As the rear foot everts, the two axes align so they are more parallel, unlocking the foot and allowing it to conform to the surface and/or absorb the ground reaction force (GRF). Conversely, as the rearfoot inverts, the two axes converge, locking the foot into a supinated position and allowing it to function as a rigid lever for propulsion.²

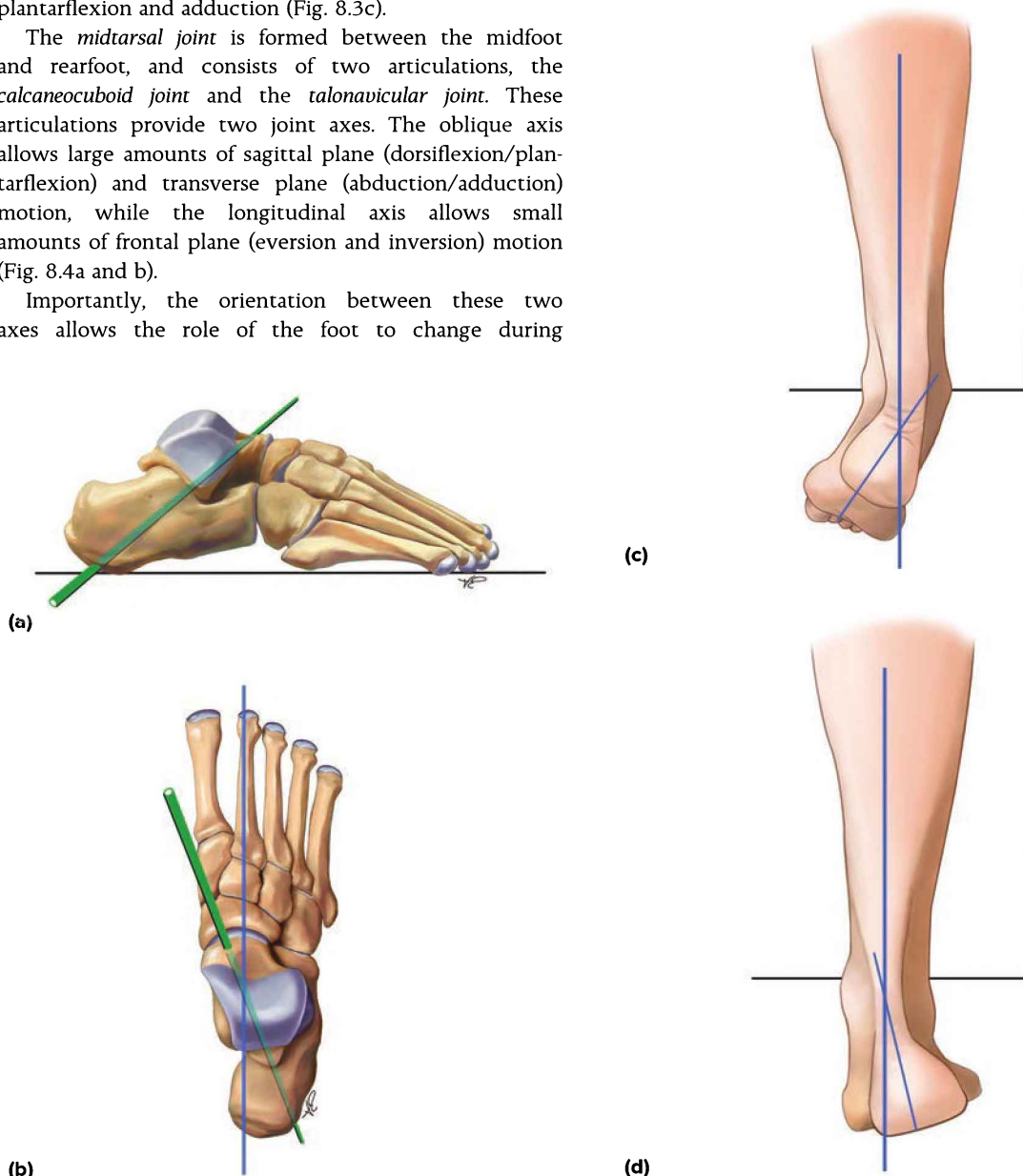
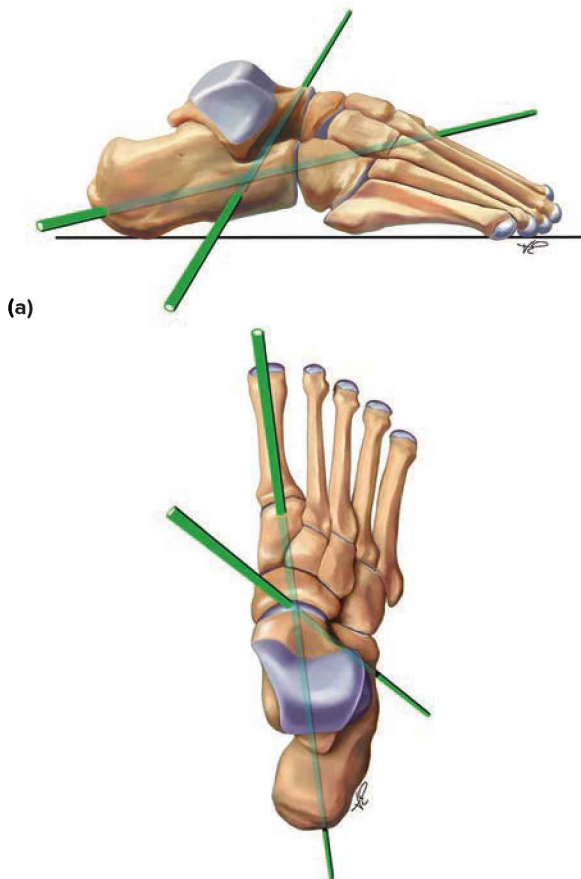


Figure 8.3 Axis of motion of the subtalar joint (a) Lateral view. Angle of inclination approximately 50° to transverse plane (b) Superior view. Angle between axis of motion of subtalar joint and longitudinal axis of the rearfoot is approximately 15° (c) Supination at subtalar joint with 20° calcaneal inversion (d) Pronation at subtalar joint with 10° calcaneal inversion



(b)

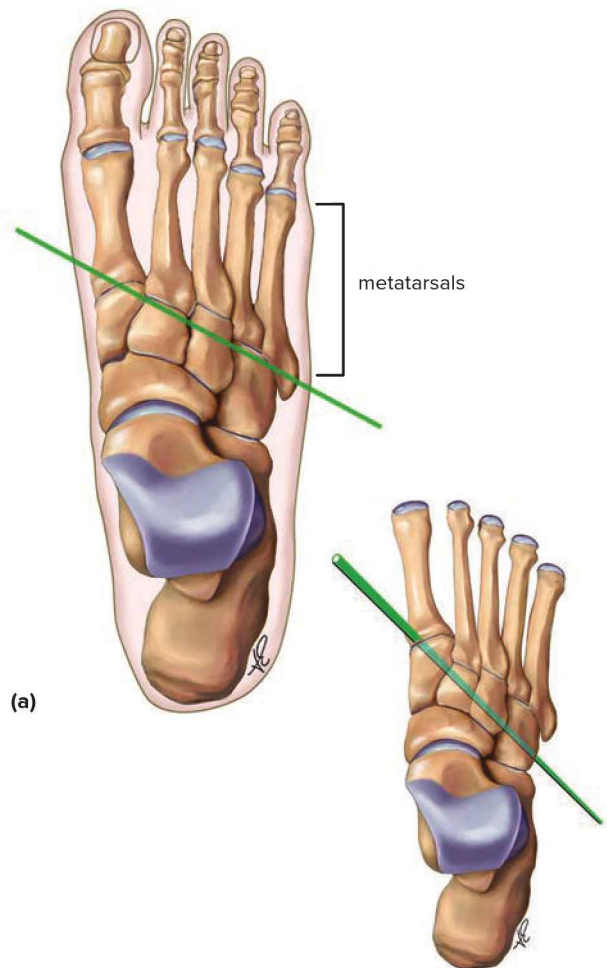
Figure 8.4 Oblique and longitudinal axis of midtarsal joint
(a) Lateral view (b) Superior view

The *midtarsal* (or *Lisfranc*) joints are formed between the distal tarsal bones of the midfoot (cuneiforms and cuboid) and the five metatarsal bones (forefoot). The axis of motion for these joints runs primarily in the transverse plane in an oblique direction (Fig. 8.5a and b). This leads primarily to sagittal plane motion (flexion/extension), although some frontal plane motion (eversion/inversion) does occur medially (adduction, Fig. 8.5c) and laterally (abduction, Fig. 8.5d).

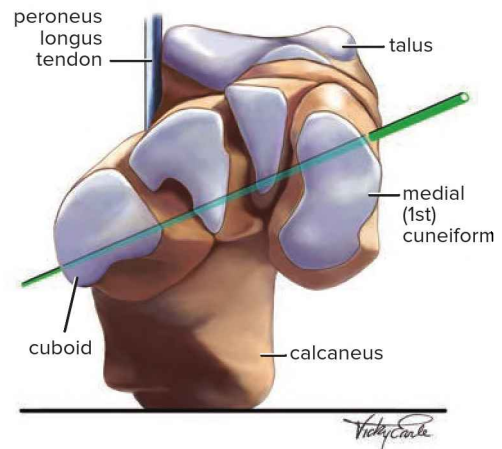
The *first metatarsophalangeal joint* (MTP) is formed between the head of the first metatarsal and the base of the proximal phalanx. The primary motion that occurs at this joint is in the sagittal plane (flexion/extension) (Fig. 8.6). In particular, extension of this joint is essential to optimise function of the windlass mechanism (see below) during gait.

Ideal neutral stance position

To examine stance position, have the patient adopt a normal, comfortable, standing posture. Ideal neutral stance occurs when the joints of the lower limbs and feet are symmetrically aligned, with the weight-bearing



(a)



(b)

Figure 8.5 (a) midtarsal joints—the joints between the distal tarsal bones of the midfoot (cuneiforms and cuboid) and the five metatarsal bones (forefoot) (b) The metatarsal break (frontal plane). The axis of motion runs as shown (green rod)

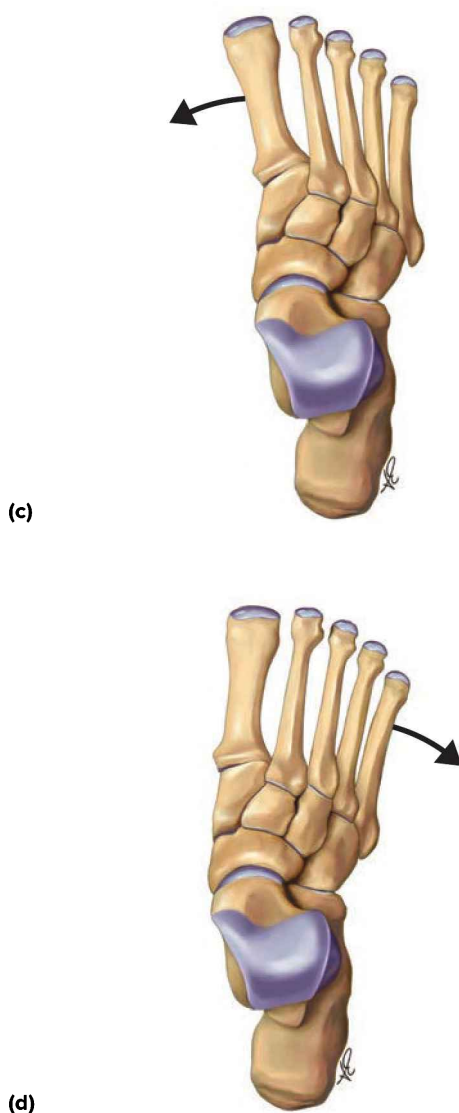


Figure 8.5 (cont.) Frontal plane motion of the metatarsal break showing that the forefoot can (c) adduct and (d) abduct

line passing through the anterior superior iliac spine, the patella and the second metatarsal (Fig. 8.7).

When the feet are in a symmetrical position, the subtalar (talocalcaneal) joint is neither pronated nor supinated, and the midtarsal joint (talonavicular and calcaneocuboid joints) is maximally pronated so that the first and second metatarsal heads are in contact with the ground. The long axis of the forefoot through the second metatarsal is perpendicular to the bisection of the heel (Fig. 8.8) and in line with the tibial tuberosity. The ankle joint is neither plantarflexed nor dorsiflexed, and the tibia is perpendicular

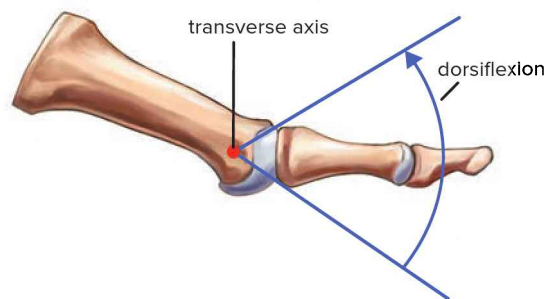


Figure 8.6 Motion of the hallux around the transverse axis of the first metatarsophalangeal joint



Figure 8.7 The alignment of the lower limb in neutral position. The weight-bearing line runs through the anterior superior iliac spine, patella and second metatarsal. The calcaneus is in line with the tibia, and the forefoot is perpendicular to the calcaneus

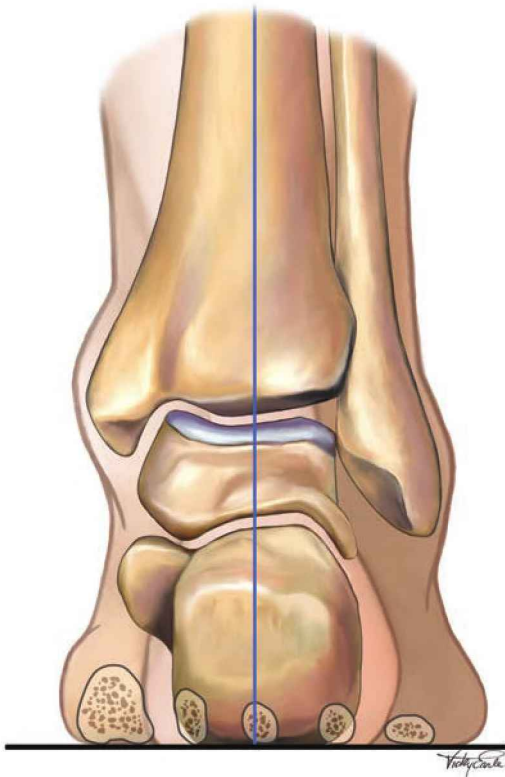


Figure 8.8 Normal relationship between the forefoot and rearfoot when the foot is in neutral stance

to the supporting surface in the sagittal and frontal plane. The knee is fully extended (but not hyper extended) and in slight valgus alignment. The hips are in a neutral position (neither internally nor externally rotated, neither flexed nor extended). The left and right anterior superior iliac spines of the pelvis are level. A slight anterior tilt of the pelvis is normal. More specific objective descriptions of ideal alignment are outlined in the assessment section.

'IDEAL' BIOMECHANICS WITH MOVEMENT—RUNNING

As injury mechanisms for many overuse injuries can be associated with theoretically suboptimal lower limb biomechanics, the clinician must know how to assess lower limb biomechanics during running. We focus first on the heel strike pattern of running as this is the predominant pattern for the majority of runners.³ We will then consider how biomechanics is altered when running with a forefoot strike pattern.

Although we focus on ideal running biomechanics, ideal walking biomechanics are similar to heel strike running patterns outlined below. The most important

biomechanical feature that distinguishes running from walking is the airborne or 'float' phase of running, where neither foot is in contact with the ground.² Additionally, vertical GRFs during running are double that of walking,^{2,4} the pelvis is in greater anterior tilt,⁵ and sagittal plane excursions of the knee and hip are increased. Ultimately, this leads to greater stress on structures of the lower limb.

The heel strike pattern of running can be split into a number of phases (Fig. 8.9), each of which will be discussed below.

Loading (heel strike to foot flat)

With the leg swinging toward the line of progression, and the foot supinated, the rearfoot (heel) contacts the ground in slight inversion (0–5°).^{2,6–8} At heel strike, the pelvis is level, in slight anterior tilt (10°), and internally rotated; the hip is externally rotated (5–10°) and flexed (20–30°); and the knee is flexed (10°). Due to the laterally directed line of the GRF produced by heel strike, a cascade of events occurs to assist shock absorption. First, the rearfoot begins to evert, accompanied by tibial and femoral (hip) internal rotation, and hip adduction. This is combined with knee flexion, which peaks at around 45° as a result of the GRF line passing posterior to the knee joint.^{9,10} Each of these motions is controlled by eccentric muscle activity which helps to dissipate the GRF. In addition, there may be contralateral pelvic drop, although this should be minimal (approximately 5°).¹⁰ The gluteal musculature should actively control this motion and further dissipate the GRF.

Initial rearfoot eversion also results in more parallel alignment of the midtarsal joints (i.e. calcaneocuboid and talonavicular), causing them to unlock.² Importantly, this allows the forefoot to make solid contact with the ground at foot flat¹¹ and allows the foot to adapt during loading to potentially uneven or unstable terrain.²

Although motions that comprise foot pronation are normal, they should not be excessive or rapid (i.e. hyperpronation). Excessive motion will place strain on structures designed to control foot pronation, such as the plantar fascia, tibialis posterior muscle and intrinsic foot musculature.² Excessive pronation also increases medial GRF, accentuating more proximal motion at the knee, hip and pelvis, and increasing load on ligamentous and muscular structures responsible for proximal control.²

The clinician should carefully note proximal motion during this early phase of stance. Excessive contralateral pelvic drop and/or hip adduction/internal rotation may increase strain on structures required to control it, such as the iliotibial band (ITB), gluteal musculature and tensor fascia latae (TFL) muscle. Additionally, this may also place increased or altered loading on the lumbar spine, tibiofemoral joint and patellofemoral joint (PFJ). Any

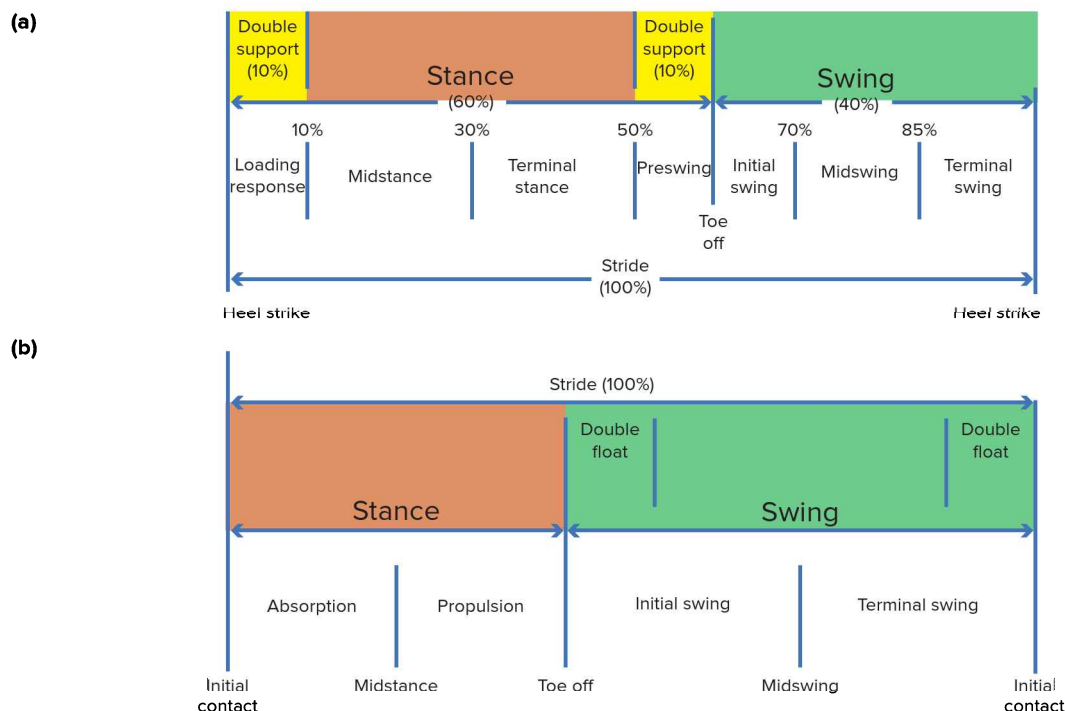


Figure 8.9 Gait cycle with phases and individual components (a) Walking (b) Running

ADAPTED FROM DUGAN AND BHAT²

excessive anterior tilting of the pelvis may place excessive strain on the lumbar spine and/or hamstring musculature, and may impair gluteal function.

Conversely, inadequate pronation or excessive supination may lead to an excessive or prolonged laterally directed GRF,² resulting in a less mobile foot and poorer shock absorption capacity. This may be associated with lower limb stress fractures,¹² or increase the incidence of lateral ankle sprain and chronic ankle instability.^{13, 14}

Midstance (foot flat to heel off)

The beginning of midstance is indicated by the forefoot making contact with the ground, normally in a neutral transverse plane position (i.e. not abducted or adducted).⁶ Lower limb biomechanical function during midstance involves a transition from biomechanics required for shock absorption following loading, to biomechanics required for propulsion. During this time, the ankle moves towards maximal dorsiflexion (approximately 20°) to allow forward motion of the tibia and the centre of mass (COM) to pass over the stance leg.¹⁵ Excessive ankle dorsiflexion can lead to increased strain on the plantar fascia, Achilles tendon, and associated gastrocnemius and soleus musculature. At the same time, the hip and knee are moving from flexion towards extended positions, assisting forward motion of the body's COM.

Maximal foot pronation followed by maximal ankle dorsiflexion should be reached immediately after the body's COM has passed anterior to the stance limb.¹¹ Peak rearfoot eversion should reach approximately 10°,^{6, 16} and peak forefoot abduction approximately 5°. The rearfoot then begins to invert and the forefoot adducts, causing the foot to supinate, and the tibia and femur to externally rotate.¹¹ This external rotation action is assisted by force transmission from the externally rotating pelvis, which results from momentum of the swinging contralateral limb.

There are a number of things for the clinician to consider during this phase. Excessive pronation, or delayed/failed transition from shock absorption to propulsion actions by the lower limb, may be detrimental to a number of structures. Firstly, this will place excessive strain on structures responsible for controlling pronation and may increase the risk of conditions such as plantar fasciopathy, Achilles and tibialis posterior tendinopathies, proximal tibial periostitis, or tibial stress fractures due to excessive pull of the tibialis posterior and long flexors. Also, continued instability of the foot may lead to development of first MTPJ abnormalities, including hallux valgus, sesamoid pain and/or excessive interdigital compression (Morton's neuroma). If left untreated,

over time this instability may also lead to metatarsal or sesamoid stress fractures.

More proximally, excessive or prolonged pronation may also result in abnormal transverse and frontal plane motion at the hip and knee due to a delay in external rotation. Ultimately, this can place excessive strain on many structures such as the PFJ, patellar tendon (both conditions discussed in Chapter 36), ITB (Chapter 37), and musculature that control this motion. Conversely, the same proximal anomalies may result due to inadequate pelvic and hip control. The source of the biomechanical dysfunction may need to be determined through further structural and functional tests (see assessment section below).

Propulsion (heel off to toe off)

Following heel off, the foot continues to supinate. Importantly, as this occurs, inversion of the rearfoot causes the transverse tarsal joint axes to converge.² This convergence of joint axes causes the midfoot to lock into position, creating a rigid lever.² Concurrently, the stance limb continues to externally rotate, the hip reaches maximal extension of between 0 and 10°,^{5, 15, 17} and the knee flexes once more due to hamstring muscle contraction.^{15, 17} Additionally, acceleration of the stance limb is provided through plantar flexion at the ankle, produced by the gastrocnemius and soleus complex.¹⁸ This same gastrocnemius and soleus activity, along with the tibialis posterior and intrinsic foot musculature activity, continues to actively assist supination of the foot, and maintain its function as a rigid lever.² Passively, rigidity of

the foot is supported by the windlass mechanism, that is, increased tension of the plantar fascia due to extension of the metatarsals, which pulls the calcaneus and metatarsal heads together (Fig. 8.10).² By toe off, the rearfoot should be inverted to approximately 10°, and the forefoot adducted approximately 5°.⁶

Failure of normal propulsion will cause an inefficient gait pattern. This can limit performance and predispose to injury for several reasons. First, the peroneal musculature may be forced to work harder to stabilise the medial and lateral columns of the foot which can lead to peroneal tendinopathy and/or stress fracture of the fibula.

Second, impaired supination may lead to toe off via the lateral rays instead of the first ray. This may compress the transverse arch of the foot excessively, and lead to interdigital nerve compression (Morton's neuroma) and risk of lateral forefoot stress fracture.

Third, reduced propulsion from the stance limb may increase reliance on the swing phase to produce forward momentum. To achieve this, the hip flexors, rectus femoris and iliopsoas will generate more rapid hip flexion, increasing the potential for tendinopathies. Additionally, to compensate for impaired propulsion, pelvic and trunk rotation may increase, leading to increased strain on spinal structures.

Initial swing

Following ipsilateral toe off, the body is thrust into the first 'float' phase where neither limb is in contact with the ground. Rectus femoris and iliopsoas muscle activity continue the forward momentum of the now swinging

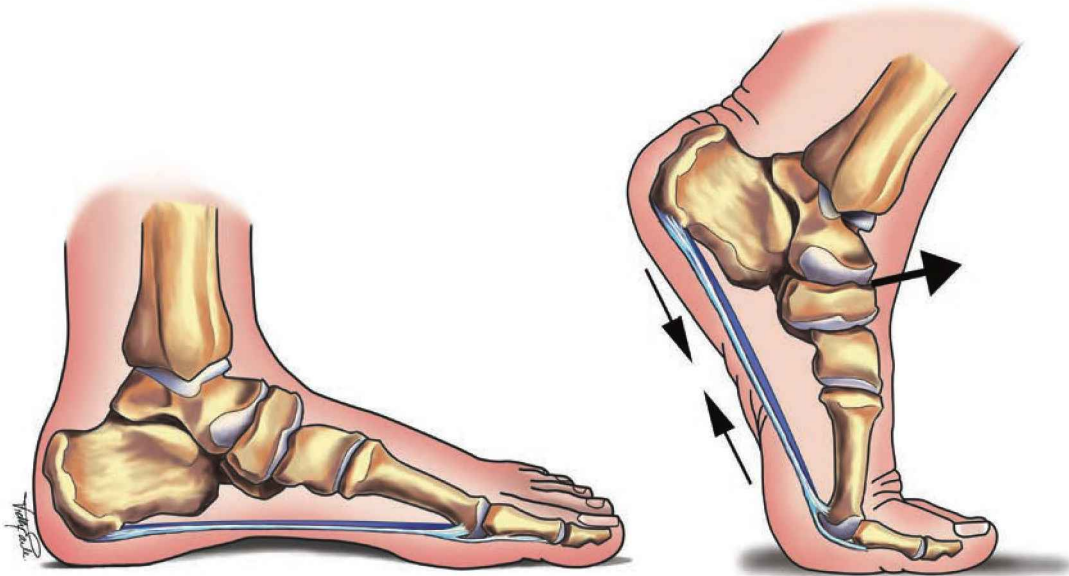


Figure 8.10 The windlass mechanism comes into play after heel off. Metatarsal extension increases tension on the plantar fascia, and forces the transverse tarsal joint into flexion which increases stability at push off

limb.¹⁹ As the limb advances, the pelvis moves with it, thrusting the hip into abduction and external rotation, which is in turn controlled by the hip adductors.¹⁹ The tibialis anterior contracts to begin dorsiflexing the foot in preparation for terminal swing.¹⁹ While these motions continue, they are aided by the addition of a new stable support when the contralateral limb strikes the ground and commences its own *loading* phase. Continuation of normal swing at this time relies on the ability of the contralateral gluteal musculature to dissipate the GRF produced by this event and prevent the pelvis dropping on the swing side. Failure to do so will increase the work required by the hip and knee flexors to clear the swinging limb, possibly leading to overuse injury.

Terminal swing

Following contralateral toe off, the body is thrust into the second 'float' phase. During this time the ipsilateral swinging hip reaches maximal flexion (approximately 30°),^{5, 9} before being brought under control by the hamstring and gluteal musculature.¹⁹ The same hamstring activity slows the rapidly extending knee in preparation for heel strike. At the same time, the hip adductors, which have been working eccentrically to control abduction of the swinging limb, begin to work concentrically to adduct the hip and bring it toward the midline.¹⁹

Angle and base of gait

The angle of gait is the angle between the longitudinal bisection of the foot and the line of progression (Fig. 8.11). The normal angle of gait is approximately 10° abducted

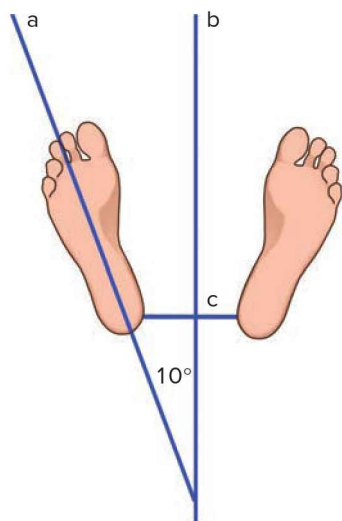


Figure 8.11 The angle of gait is the angle between (a) the long axis at the foot and (b) the line of progression. (c) Base of gait is the distance between the medial aspects of the heels

from the line of progression in walking. Abducted gait describes an angle of gait greater than 10°. The angle of gait reflects the hip and tibial transverse plane positions. The base of gait is the distance between the medial aspect of the heels (Fig. 8.11). A normal base of gait is approximately 2.5–3 cm.

Changes from the normal angle and base of gait may be secondary to structural abnormalities or, more commonly, as compensation for another abnormality. For example, a wide base of gait may be necessary to increase stability. As running velocity increases, the angle and base of gait decrease. With faster running, the angle of gait approaches zero and foot strike is on the line of progression. This limits deviation of the COM as the lower limbs move beneath the body, thus allowing more efficient locomotion (Fig. 8.12).

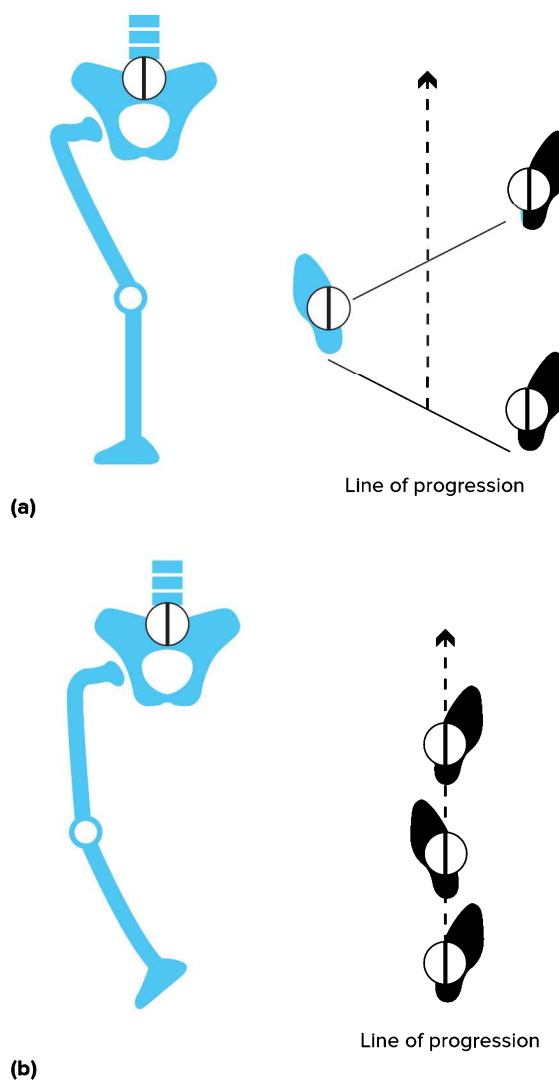


Figure 8.12 Angle and base of gait (a) Walking (b) Running

Landing point relative to centre of mass

Greater distance between foot strike point and COM (i.e. overstride) has been reported to increase lower limb joint loads,²⁰ and as a result is thought to increase the risk of running injury development. When measuring this clinically, we suggest this distance should not exceed more than one third of a foot length (Fig. 8.13), with greater than a foot length considered a marked overstride. During sprinting, the foot should land almost directly under the COM.

Influence of gait velocity

Increased gait velocity influences a number of biomechanical factors. As gait velocity is increased, greater emphasis is placed on the swinging actions of the upper limbs, trunk and lower limbs to produce forward momentum.² This difference has significant implications for the flexibility and eccentric muscle control requirements of these structures (e.g. ipsilateral hamstrings strain during late swing). Greater excursion of the proximal joints (knee, hip and pelvis) also occurs with increased velocity, placing increased reliance on eccentric muscle control.^{19, 21} At the foot and ankle, the bones making up the rearfoot, midfoot and forefoot all

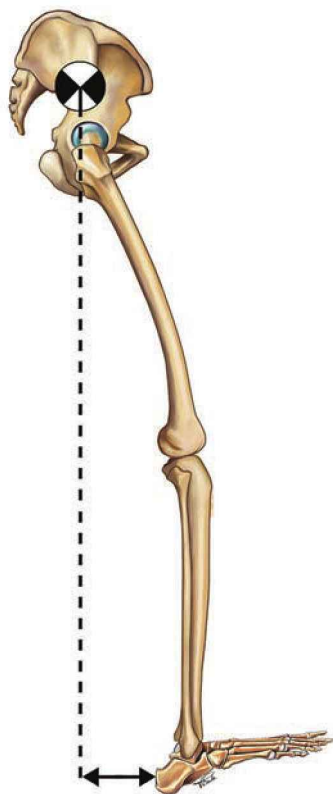


Figure 8.13 Heel to COM distance at initial contact during running²⁰

reduce their excursions in all three planes with increased velocity.²² This indicates the need for stiffer joint structures with increased speed and a greater demand on intrinsic foot musculature control.²²

In slower running, the stance phase takes longer than the swing phase. As running speed increases, stance phase and flight phase times approach each other until the stance phase becomes shorter than the swing phase in sprinting (Fig. 8.14).²

Additionally, as running velocity increases, foot strike patterns may be altered. As mentioned previously, foot strike patterns are similar between slow running and walking for most individuals (Fig. 8.15). During faster running (striding), the foot may strike with the heel and forefoot simultaneously prior to heel off (midfoot strike), or may strike with the forefoot initially followed by heel lowering to the surface prior to heel off (forefoot strike). In sprinting, weight-bearing is maintained on the forefoot from contact to toe off, although the heel may lower to the supporting surface at midstance. In some individuals, this pattern can commence even at slower running speeds, or immediately upon initiation of a run. In particular, some habitual barefoot runners often have a natural forefoot strike pattern regardless of velocity.³

Comparing heel and forefoot strike patterns

Changing from a heel strike to a forefoot strike pattern has significant implications for lower limb biomechanics and assessment. This has become particularly important

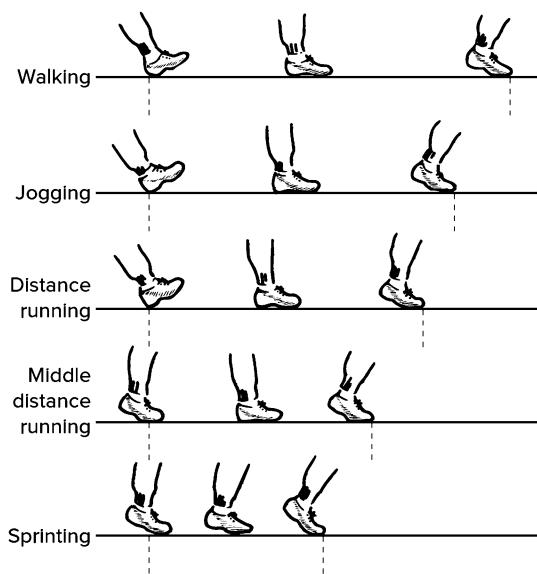


Figure 8.14 Pattern of the stance phase during different speeds of walking and running

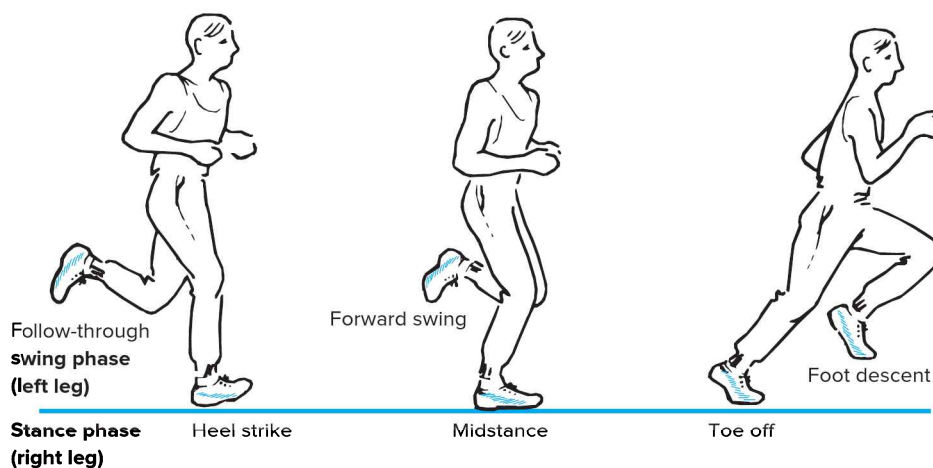


Figure 8.15 The swing and stance phases of running

to clinical practice with the recent popularity of *Born to Run*²³ and many runners choosing to transition from a habitual heel to a forefoot strike. Firstly, forefoot strike patterns result in slight plantar flexion of the foot at impact, followed by dorsiflexion as the heel lowers to the ground.² This means that the ankle joint is more compliant and able to absorb GRF in the sagittal plane, and can lead to a reduction in vertical GRF and loading rates following foot strike.^{3, 6} Additionally, forefoot striking has also been reported to reduce knee and hip joint loads, and these changes are often used to justify transition toward a forefoot strike for hip and knee injuries.²⁴ However, concurrent increases to joint loads at the ankle indicate potential detrimental increases to tissue stresses distally,²⁴ which fits with many anecdotal reports of Achilles tendinopathy, plantar fasciopathy and metatarsal stress fractures as a result of transitioning from heel to forefoot strike.

Influence of fatigue on running biomechanics

Some runners will report pain that only occurs following prolonged activity. For example, an individual with patellofemoral pain (Chapter 36) may report no pain during the early stages of a run, but cease running due to severe pain after five kilometres. This can make clinical assessment difficult, since the condition may result from suboptimal biomechanics that only occur with fatigue. For example, excessive hip adduction during running in the individual with patellofemoral pain may occur due to poor gluteal muscle endurance. Therefore, the clinician should evaluate functional biomechanics both at baseline and following fatigue and/or onset of pain. In the clinical setting, this means scheduling the appointment so that the patient can be seen before and immediately after a run, or having them run to a fatigued state before attending the clinic.

LOWER LIMB BIOMECHANICAL ASSESSMENT IN THE CLINICAL SETTING

This section aims to help the junior clinician develop a routine for efficient lower limb biomechanical assessment. There is no single best way to assess biomechanics and the experienced clinician will vary his or her assessment depending on the clinical presentation.

For this example, consider the patient to have patellofemoral pain—a condition that warrants assessment of the entire kinetic chain.

PRACTICE PEARL

Two guiding principles will help with the efficiency of the office assessment and ensure it is appropriately comprehensive.

1. Examine from distal to proximal (start at the foot and then examine proximally to the pelvis and trunk).
2. Examine the patient in 'static stance' first before increasingly challenging him or her with 'functional' tests before moving to fully 'dynamic' or 'sport-specific' tests as appropriate (see Fig. 8.16 for the concept of the hierarchy). These terms are explained below.

Thorough biomechanical assessment may require the patient to stand, walk, run, land on two feet and land on one leg only. Assessment during 'function' (playing sports, executing certain sporting activities such as a kick or a pirouette) may also be relevant.

We explain each of these steps in the order that many experienced clinicians perform the assessment. The major elements in the assessment are:

- structural (static) biomechanical assessment (Fig. 8.17a-g)
- functional lower limb tests—single-leg stance, heel raise, squat and landing (Fig. 8.18a-g)

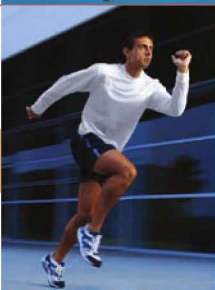
	POSITION	ACTIVITY
Static stance		A good starting place. The patient stands in a comfortable position, weight centrally distributed over both feet (see also Fig. 8.17a–g).
Simple functional		This refers to a group of simple movements in isolation—single-leg standing, single-leg heel raise, single-leg squat, step down, hopping and landing (see also Fig. 8.18a–g).
Dynamic movements		This refers to activities such as running. This may need to be done outside the office on an adjacent track, park or car park depending on what is available.
Sport-specific activity		If the athlete is a basketball player, ballet dancer, or pole vaulter (for example), the clinician may also need to observe the athlete performing sporting activities that are relevant to the presenting complaint.

Figure 8.16 An overview to guide lower limb biomechanical assessment (See also Figures 8.17 and 8.18.)

- assessment of a patient's running biomechanics ('dynamic assessment')
- detailed sport-specific tests as indicated by the above tests and the clinical presentation.

Structural ('static') biomechanical assessment

The clinician performs the assessment of static stance by critically viewing the foot, ankle, knee and pelvis (Fig. 8.17a).

Foot

Inspect the foot subjectively (does it look abnormal?), quantify posture using the six item Foot Posture Index (FPI) below and also pay attention to first MTP range of motion (Fig. 8.17b).

Foot Posture Index

Quantify standing foot posture objectively with the FPI, a test that reflects foot posture at the rearfoot, midfoot and forefoot, as well as an overall impression of foot



(a)



(b)

Figure 8.17 Static assessment of the lower limb (a) With the patient in this comfortable position, static stance examination begins at the foot. The examiner then assesses the ankle, knee and hip/pelvis (b) Jack's test for first MTPJ dorsiflexion range of motion. Normal range of motion is approximately 50° relative to the floor

The Foot Posture Index ²⁶—a rapid, quantitative measure of static foot biomechanics

Each of the six items in the FPI is scored as -2 (highly supinated), -1 (supinated), 0 (neutral), $+1$ (pronated) or $+2$ (highly pronated); this leads to sums between -12 (highly supinated) and $+12$ (highly pronated).²⁶

Talar head palpation

The talar head is palpated on the anterior aspect of the ankle. If the head can be felt equally on the medial and lateral side, a neutral score (0) is given. If greater prominence is felt medially, a pronated score is given ($+2$ defined by only medial prominence felt) and if greater prominence is felt laterally, a supinated score is given (-2 defined by only lateral prominence felt).

Supra and infra lateral malleolar curvature

A neutral score for this item is given if the curves above and below the lateral malleolus are equal when viewed posteriorly. If the curve above the malleolus is flatter, a pronated score is given ($+2$ defined by completely flat) and if the curve below the malleolus is flatter, a supinated score is given (-2 defined by completely flat).

Calcaneal frontal plane position

A neutral score is given if the rearfoot is perpendicular to the floor. A more valgus rearfoot relative to the floor is given a pronated score ($+2$ defined by $>5^\circ$) and a more varus rearfoot relative to the floor is given a supinated score (-2 defined by $>5^\circ$).

Bulging in the region of the talonavicular joint (TNJ)

A neutral score is given if the skin immediately superficial to the TNJ is flat. If the TNJ is bulging, a pronated score is given ($+2$ defined by marked bulging) and if the TNJ area is concave (indented) a supinated score is given (-2 defined by marked concavity).

Height and congruence of the medial longitudinal arch

A neutral score is given if the arch shape is uniform and similarly shaped to the circumference of a circle. If the arch is flattened and lowered, a pronated score is given ($+2$ defined by the mid-portion of the arch making contact with the floor). If the arch is high, a supinated score is given (-2 defined by an acutely angled posterior end of the arch).

Abduction/adduction of the forefoot on the rearfoot

A neutral score is given when the forefoot can be seen equally on the medial and lateral side when viewed from behind the axis of the rearfoot. If more of the forefoot is visible laterally than medially, a pronated score is given ($+2$ defined by only lateral forefoot visible) and if more of the forefoot is visible medially than laterally, a supinated score is given (-2 defined by only medial forefoot visible).

The total FPI score for the normal healthy population is 2.4 (i.e. slightly pronated).²⁸ Thus, scores of 0 to $+5$ are considered neutral. A score of $+6$ to $+9$ is considered pronated, $\geq +10$ highly pronated, -1 to -4 supinated and -5 to -12 highly supinated.

type. It requires no equipment and takes approximately 2 minutes to complete.^{25, 26, 27}

Additional background information about the FPI, including definitions and pictures of various foot types for each item, are shown in the user guide and manual.²⁵ Note that, as with any clinical skill, training and experience are important. The clinician should rate at least 30 individuals with a broad range of foot types before applying the FPI clinically.^{29, 30, 31}

Jack's test for first metatarsophalangeal joint (MTPJ) range (plantar fascia integrity)

The clinician can rapidly assess the first MTPJ and also the integrity of the plantar fascia using 'Jack's test' (Fig. 8.17b).³² The normal range of first MTPJ dorsiflexion motion should be around 50° relative to the floor. As the first ray dorsiflexes, tightening of an intact plantar fascia should cause the rearfoot to invert. If the rearfoot does not move, it suggests poor plantar fascia integrity which can result in inadequate supination during the propulsive phase of gait. Additionally, increased resistance to or a reduction in

motion may result from the presence of a valgus aligned forefoot.³³ Both these structural issues can be corrected using orthoses, taping and/or corrective exercise.

Ankle dorsiflexion

Accurate ankle dorsiflexion in weight-bearing using an inclinometer provides a more useful measure of range of motion at this important joint than does a rough assessment when the patient is lying on a treatment couch. This ankle biomechanical assessment is best done with the knee both extended and flexed (Fig. 8.17c and d). Normal ankle dorsiflexion range with the knee flexed is 45° and with the knee extended is 40°.³⁰ We recommend that clinicians have an inclinometer readily available (Fig. 8.17c and d)—it adds a great deal of accuracy to measurements and reduces assessment time.

An alternative way of quickly assessing weight-bearing ankle dorsiflexion is to have the patient perform the 'knee-to-wall' test (Fig. 8.17e). This provides an efficient outcome measure for ankle dorsiflexion range when gastrocnemius tightness is not a concern and this can be

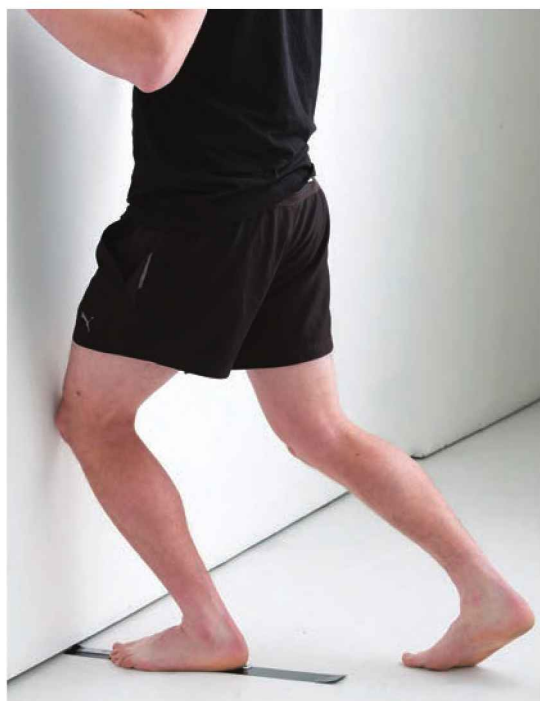


(c)



(d)

Figure 8.17 (cont.) (c) Assessing ankle dorsiflexion in weight-bearing with the knee extended (d) Assessing ankle dorsiflexion in weight-bearing with the knee flexed



(e)

Figure 8.17 (cont.) (e) The knee-to-wall test. It is useful to have a line on the floor that continues up the wall. The patient lines up their second toe and mid-calcaneus on the horizontal line and lunges their patella toward the vertical line. This minimises the likelihood of going into pronation to achieve greater range of motion (ROM). The distance from the wall to the second toe is recorded

PRACTICE PEARL

If ankle dorsiflexion range differs in knee extension and flexion by more than 5° , it suggests there is limitation of dorsiflexion with the knee extended; this points to gastrocnemius tightness. If foot pronation is excessive to achieve normal ankle dorsiflexion range (i.e. 45°) with the knee bent, excessive pronation may be occurring during functional activities to achieve full functional movement (e.g. when running).

particularly useful following lateral ankle sprain. With the knee-to-wall test, clinicians can determine whether there are side-to-side differences or if ankle dorsiflexion is restricted. 'Normal' values will vary depending on the size of the patient and their sport-specific demands, but, as a general rule of thumb, the distance from the wall to the second toe should be greater than 8–10 cm.

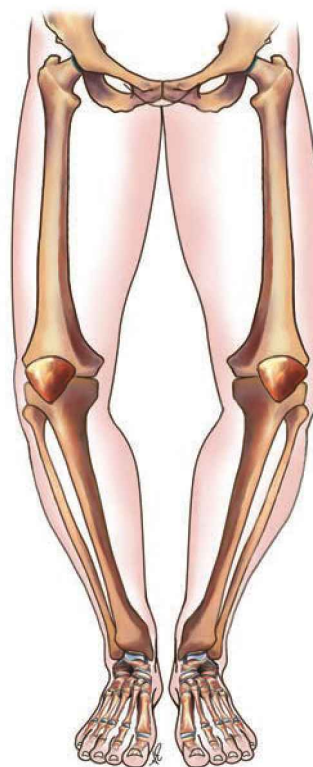
Assessment of tibiofemoral alignment at the knee

Tibiofemoral joint alignment may reflect genu varum or valgum (Fig. 8.17f and g). This can also be measured using an inclinometer, which compares favourably to the gold standard radiographic measure.³⁴

Leg length assessment

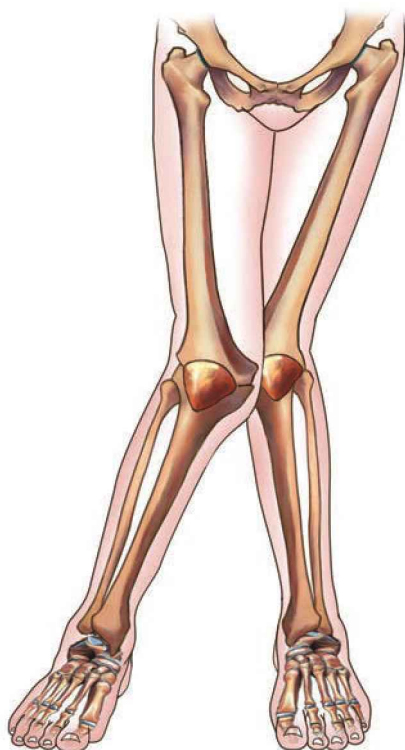
Discrepancies in leg length are extremely common, existing in up to 70% of the normal population.³⁵ Nevertheless, leg length discrepancy has long been thought to be a risk factor for a range of musculoskeletal conditions, including low back pain, hip and knee osteoarthritis, and lower limb stress fractures.^{35, 36}

Laboratory studies have shown that leg length differences alter the distribution of mechanical stress within the body, particularly by increasing the magnitude of forces through the shorter limb.³⁷ However, prospective risk-factor studies in athletes are lacking and what constitutes a clinically significant leg length difference remains unknown. Subsequently, the extent to which leg length differences should be



(f)

Figure 8.17 (cont.) Static assessment of the lower limb—alignment at the knee (f) Genu varum



(g)

Figure 8.17 (cont.) (g) Genu valgum

corrected in athletes remains a source of controversy, with opinions ranging from 3 mm to 22 mm.³⁸ We recommend that clinicians take a 'treatment direction test' approach (see below) and adopt corrections if they substantially change symptoms.

Leg length can be measured in a variety of ways, including clinical measures using a tape measure in standing (anterior superior iliac spine or ASIS to floor) and in supine (ASIS to medial malleolus). However, these methods may be inaccurate, with reports of poor inter- and intra-rater reliability and errors of up to 20 mm compared to radiographic imaging.³⁸ There are a number of radiographic techniques to evaluate leg length differences, including magnetic resonance imaging (MRI), computed tomography (CT) 'scanograms' and standing full leg radiograph. Each of these methods has demonstrated high accuracy with average errors of less than 2 mm.³⁹

Summary of static assessment

The 'static' biomechanical assessment of the foot, ankle and knee provides a substantial amount of valuable clinical information and can be completed within 5 minutes.

Possible mechanisms that underpin common clinical observations are shown in Table 8.2.

Functional lower limb tests

The next step is to assess simple functional movements. The patient should do these tests both with and without sporting footwear where appropriate. This will help the clinician determine whether the individual's footwear is detrimental, beneficial or has no effect on functional biomechanics.

Single-leg stance with progressions

The single-leg stance test begins to challenge lower limb balance and proprioception (Fig. 8.18a). Inability to maintain the single-leg stance position is likely to carry over to suboptimal biomechanics during sporting activity. The patient performs this test with (i) eyes open, (ii) eyes closed and (iii) challenged further by also performing a single heel raise. Depending on the balance requirements for the individual, once balance can be maintained in single-leg stance for at least 10 seconds, balance can be assessed using more challenging activities, including variations in surfaces and ability to adapt to internal and external perturbations.



(a)

Figure 8.18 Functional assessment of the lower limb.
(a) Single-leg stance to evaluate alignment and control

Table 8.2 Common lower limb biomechanical observations, possible mechanisms and confirmatory assessments

Observations	Possible mechanisms	Confirmatory assessments
Excessive or asymmetrical pelvic or trunk movement (frontal, transverse, sagittal planes)	Inadequate ROM (hip)	Clinical ROM tests with inclinometer (Chapter 31); figure '4' test
	Inadequate strength (abdominals, lumbopelvic muscles, hip abductors)	Manual muscle tests
	Altered neuromotor control (hip abductors, lumbopelvic muscles)	Biofeedback
	Decreased muscle length (hamstrings, rectus abdominis, rectus femoris)	Muscle length tests
	Lumbar spine/sacroiliac joint stiffness/pain	Joint palpation
Increased hip adduction/femoral internal rotation	Structural (femoral anteversion)	Radiographic: MRI, X-ray Clinical assessment
	Inadequate ROM (hip)	Clinical ROM tests with inclinometer (Chapter 31); figure '4' test
	Inadequate strength (hip external rotators, abductors)	Manual muscle test (Chapter 31) or hand-held dynamometry
	Altered neuromotor control (hip external rotators, hip abductors)	Biofeedback
Increased apparent knee valgus	Structural (genu varum, tibial varum, coxa varum)	Imaging: MRI, X-ray Clinical assessment: goniometer, inclinometer
	Inadequate ROM (hip)	ROM tests: clinical (Chapter 31) inclinometer; figure '4' test
	Inadequate strength (hip external rotators, hip abductors, quadriceps, hamstrings)	Manual muscle test (Chapter 31) or hand-held dynamometry Active gluteal and tensor fasciae latae trigger points
	Altered neuromotor control (hip external rotators, hip abductors, lumbopelvic muscles)	Biofeedback Active gluteal and TFL trigger points
Ankle equinus	Inadequate ROM (ankle)	ROM tests
	Tight gastrocnemius or soleus	ROM tests with knee flexed and extended (Fig. 8.17c and d)
Excessive or prolonged foot pronation	Pronated foot type	FPI
	Impaired windlass mechanism	Jack's test (Fig. 8.17b)
	Tibialis posterior weakness	Single-leg heel raise; manual muscle test; inability to form arch
	Ankle equinus	Ankle dorsiflexion measures
Excessive or prolonged foot supination	Leg length discrepancy (structural or functional)	Clinical measurement and radiographic imaging
	Supinated foot type	FPI
	Chronic ankle instability	Ankle ligament integrity tests
	Leg length discrepancy (structural or functional)	Clinical measurement and radiographic imaging
Reduced propulsion	Impaired windlass mechanism	Jack's test (Fig. 8.17b)
	Tibialis posterior weakness	Single-leg heel raise; manual muscle test; inability to form arch
	Pronated foot type	FPI

ROM = range of motion; FPI = foot pronation index

Single-leg heel raise (with a focus on tibialis posterior function)

Tibialis posterior is an under-recognised contributor to normal lower limb biomechanics—it is particularly important for control of foot pronation and helps stabilise the plantar arch during activity. Through its attachments to the navicular, cuneiforms, cuboid and bases of the second to fourth metatarsals, tibialis posterior inverts the subtalar joint. It is a primary dynamic stabiliser of the foot against eversion forces and is also important for propulsion.

A simple functional test, the *single-leg heel raise*, tests the ability of tibialis posterior to re-supinate the foot during propulsion of gait (Fig. 8.18b). Tibialis posterior muscle weakness will manifest as an inability to rise up through the medial aspect of the foot and invert the rearfoot toward the end of the heel raise. (Note that the same procedure can be prescribed as a therapeutic exercise when deficits are observed—this may initially require the use of support or starting in bilateral stance).

If tibialis posterior problems or excessive dynamic foot pronation are suspected, the ‘arch form’ test can be used

to evaluate the capacity of the intrinsic muscles of the foot and tibialis posterior to control the foot during stance in gait. The patient is instructed to gently lift up the inside arch while pushing the first metatarsophalangeal joint into the ground. The clinician can monitor performance by placing a finger underneath the joint to ensure sufficient downward pressure. The patient should be able to maintain this for 10 seconds. As above, the procedure provides a therapeutic exercise when deficits are observed. Note that the arch form test would not necessarily be part of a routine rapid biomechanical assessment—it provides additional information if this is warranted (e.g. if the patient demonstrates excessive or rapid foot pronation).

Single-leg squat to assess knee, hip and trunk muscle function

The continuum of activities from single-leg squat (at approximately 45° knee flexion), step down, hopping, to landing, provides a logical progression to the lower limb biomechanical assessment. There are many variations for performing a single-leg squat, including depth (knee flexion angle), arm position (crossed, hands on hips, no constraints) and posture of the unsupported leg (in front, behind) (Fig. 8.18c and d). Clinicians should ensure that they always use the same technique.

Assessment of hip and trunk function

Five main observations may indicate altered hip or trunk muscle function. The first four can be observed from in front of the athlete and the fifth is an overall assessment.⁴⁰

Trunk

Trunk lean (and/or rotation) toward the stance leg may be an adaptation to altered control of hip abduction/rotation or trunk lateral flexion/rotation. This may be observed as a more lateral position of the shoulder, relative to the hip (Table 8.2, Fig. 8.18d).

Pelvis and hip

Altered control of hip abduction/rotation or trunk lateral flexion/rotation may present as either (a) inability to maintain a level pelvis or (b) ipsilateral shift of a level pelvis. Both presentations may be observed as a lateral hip (ASIS), relative to the knee (hip adduction) and may also be referred to as a ‘Trendelenburg sign’ (Table 8.2, Fig. 8.18d).

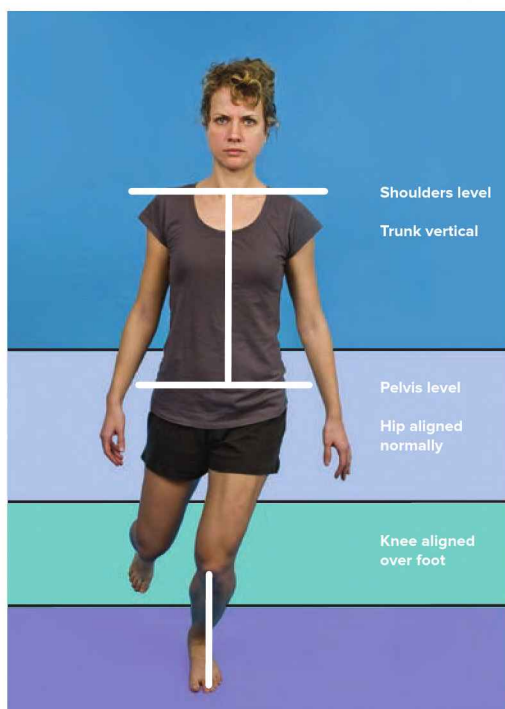
Knee

Does the centre of the knee remain over the centre of the foot? If the knee deviates toward a more medial position (relative to the foot), this is an indication of increased

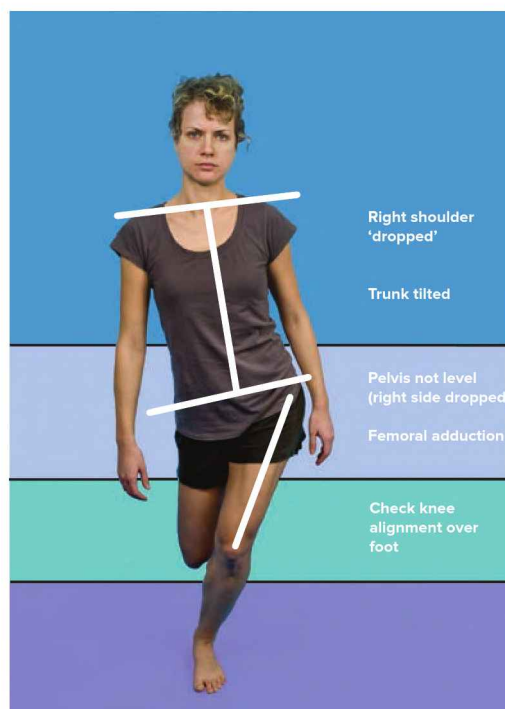


(b)

Figure 8.18 (cont.) (b) Single-leg heel raise. Look for signs of tibialis posterior weakness, including (i) inability to rise up through the medial arch or (ii) failure of the heel to invert



(c)



(d)

Figure 8.18 (cont.) Single-leg squat (c) Good form (d) Poor form

hip internal rotation and/or adduction and appears as an apparent knee valgus posture. Increased hip internal rotation/adduction may result from altered control of hip muscles (e.g. hip external rotators) (Table 8.2).

Overall impression

An individual with altered hip/trunk muscle function may exhibit global signs, such as poor quality of movement/coordination, inability to squat to the desired depth, or inability to control speed, depth and balance.

Landing—specific considerations

There are a number of ways to evaluate single-leg and double-leg landing biomechanics in the clinical setting. The clinician should determine what is appropriate taking into account the individual's sporting requirements and injury history. It may be more appropriate to evaluate single-leg landing if a higher-functioning athlete presents with a lower limb injury. The single-leg landing may be the best way to identify biomechanical deficiencies such as increased knee abduction, and decreased knee, hip and trunk flexion.

Conversely, in individuals recovering from injury and/or surgery, double-leg landing may be a more appropriate

test so that healing tissue does not receive excessive stress. It may also be important to evaluate landing performance both pre- and post-exercise, since fatigue is associated with increased knee abduction and reduced knee flexion during landing.^{41, 42}

We suggest evaluating both double-leg and single-leg landing using a drop landing assessment from a 30 cm high platform (Fig. 8.18e–g). The clinician should observe the landing pattern for signs of reduced knee and hip flexion and/or an abnormally erect landing posture. Apparent knee valgus is another movement to observe. Maximum knee valgus should reach approximately 10° for females and 5° for males for both tests from this height.⁴³ Excessive valgus and/or the presence of a heavy landing pattern involving minimal knee, hip and trunk flexion may increase the risk of future knee injury or re-injury, such as non-contact anterior cruciate ligament injuries.

As with running and squatting assessment, video footage can be used for more in-depth analysis of double-leg and single-leg landing. The clinician can use slow-motion replay and computer software packages to gain a more accurate picture of the degree of knee valgus during landing.



(e)



(g)



(f)

Dynamic movement assessment (e.g. running biomechanics)

A key to clinical biomechanical assessment is careful observation of functional movement. Running is a component of many sports and the clinician should have an effective method to assess for biomechanical problems associated with running. As a clinician, look for obvious deviations from the ideal running pattern, and use this to guide further assessment and treatment decisions. Common deviations and possible implications to injury are outlined in the 'ideal' biomechanics with movement–running' section. If possible, observe athletes participating in their sport. If necessary, sport-specific skills can be broken down into component movements to simplify observation in the clinic. Furthermore, functional clinical tests outlined in this chapter may provide insight into biomechanics during more sport-specific tasks when they cannot be evaluated in the clinical setting. For example, excessive

Figure 8.18 (cont.) (e) Starting position for single-leg landing (f) View from the front (g) View from the side

hip adduction during the completion of a single-leg squat may be indicative of excessive hip adduction during running and landing.

To detect sub-optimal biomechanics with the naked eye takes years of training and experience. Video analysis can provide valuable information from multiple views and assist the clinician greatly. This is usually done by having the patient run on a treadmill. Reflective markers can be added to identify anatomical landmarks and bony alignment. Video footage can then be slowed on a replay and this may reveal otherwise hidden abnormalities. A number of freely available computer software programs can also assist with this analysis. Examples include Hudl and Kinovea. The source of any biomechanical abnormalities may be further investigated by a thorough clinical assessment.

Sport-specific assessment

A detailed sport-specific assessment may help identify factors causing or contributing to injury. It is therefore important that clinicians understand the normal biomechanics of their patient's sport, as well as the normal range of variability between athletes. We encourage clinicians to collaborate closely with coaches and sport scientists when performing sport-specific assessments.

Sport-specific biomechanical issues thought to be associated with injury are examined in Chapter 9 for a range of different sports.

Summary of the lower limb biomechanical assessment

To iterate how we opened this section, there is no single way to perform the lower limb biomechanical

assessment—it will vary by clinical specialty (e.g. physiotherapy, podiatry, medicine, exercise physiology). Also, the clinical problem will influence the order of the assessment and the relative emphasis on various elements. In this introductory chapter, we have focused on general lower limb biomechanics. This would apply, for example, to a patient with patellofemoral pain. The assessment may differ for other conditions.

CLINICAL ASSESSMENT OF FOOTWEAR—THE FOOTWEAR ASSESSMENT TOOL

Footwear assessment is a vital component of the lower limb biomechanical evaluation. The Footwear Assessment Tool is a freely available six item template to guide the clinician when assessing footwear.⁴⁴ The tool focuses on general structure and fit (Table 8.3), motion control properties and wear patterns.

Motion control is particularly important for excessive pronators. Footwear properties that influence motion control include heel counter stiffness, midfoot torsional and sagittal stability, and type of fixation (e.g. lacing). These properties can be quantified using the motion control properties scale outlined below (Table 8.4).⁴⁴ Scores range from 0 to 9, with 9 indicating the highest level of motion control.

The wear pattern of a shoe can provide insight into the biomechanics of gait. Medial tilt of the upper, medial compression of the midsole and greater medial than lateral wear of the outsole (Fig. 8.19) indicate excessive pronation. Lateral tilt of the upper, lateral compression of the midsole, and greater lateral than medial wear of the outsole reflect excessive supination.

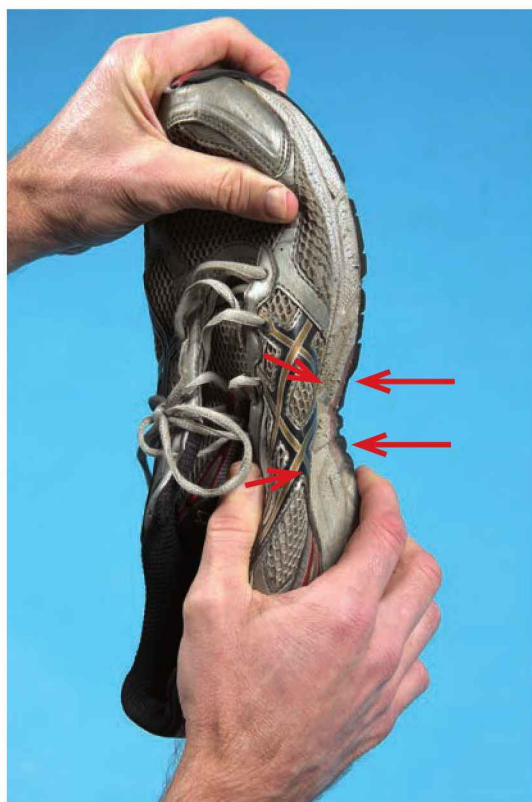
Table 8.3 Footwear assessment—general structure and fit

Item	Measurement	Potential issues
Fit	Inadequate width or depth	Joint compression (e.g. Morton's neuroma) Restriction of normal foot function
Pitch	Difference between heel height and forefoot height (typical range 0–14 mm)	Small pitch may not be suitable for someone with ankle equinus Large pitch may increase likelihood of overstride and impair attempts to transition to midfoot or forefoot strike
Last	Straight last (0–5°) Curve last (> 15°)	Accommodates a pronated foot Accommodates a supinated foot
Forefoot sole flexion point	Should line up with the first MTPJ	If too proximal, shoe stability may be impaired. If too distal, normal sagittal plane motion of the first MTPJ may be impaired, which can be problematic in the presence of poor intrinsic foot and tibialis posterior strength and function

Table 8.4 Motion control properties scale⁴⁴

Item	Score			
	0	1	2	3
Fixation (upper to foot)	None	Alternative to laces (e.g. strap, Velcro, zip)	Laces (at least 3 eyelets)	
Heel counter stiffness	No heel counter	Minimal	Moderate	Rigid
Midfoot sagittal stability	Minimal	Moderate	Rigid	
Midfoot torsional stability	Minimal	Moderate	Rigid	

Minimal – $>45^\circ$; moderate – $<45^\circ$; rigid – $<10^\circ$



(a)



(b)

Figure 8.19 Shoe wear patterns—running shoe (a) with medial compression of the upper (b) With excessive pronation, there is greater wear on the medial than the lateral sole

CONDITIONS RELATED TO SUBOPTIMAL LOWER LIMB BIOMECHANICS

Conventional wisdom has linked suboptimal lower limb biomechanics with various injuries, but there is a lack of prospective empirical evidence to confirm that biomechanical factors increase risk for lower limb

injuries.⁴⁵ Thus, for now, most biomechanical ‘risk factors’ have only level 3 to 5 evidence (Chapter 2).

Table 8.5 lists common lower limb injuries, common clinical considerations and evidence to support these considerations. It contains the ‘best available evidence’ at March 2016.

Table 8.5 Best available evidence for the association between common lower limb overuse injuries and biomechanics (as at October 2016)

Injury/condition	Associated lower limb biomechanical risk factor
Sesamoiditis	Supinated foot type ^{46, 47} Pronated foot type Forefoot valgus, plantar flexed first ray ⁴⁷ Decreased ankle DF/ankle equinus ⁴⁷ Limited first ray range of motion
Plantar fasciopathy	Pronated foot type ^{48–50} Decreased ankle DF/ankle equinus ^{49, 51, 52} Increased ankle DF ⁴⁸ Supinated foot type ⁵³ Leg length difference ⁵³
Achilles tendinopathy	Increased subtalar joint inversion ⁵⁴ Increased ankle DF ⁵⁵ Decreased gastrocnemius length ⁵⁴ Supinated foot type ⁵⁶ Pronated foot type ^{56, 57}
Lateral ankle sprain, chronic ankle instability	Prolonged pronation ⁵⁸ Decreased ankle DF ⁵⁹ Increased first MTPJ extension ^{58, 60} Increased ankle inversion at heel strike ^{13, 14}
Peroneal tendinopathy	Supinated foot type ⁶¹
Medial shin pain (medial tibial stress syndrome, shin splints)	Pronated foot type ^{62–64} Increased ankle PF ⁶⁵ Supinated foot type ⁶⁶ Decreased ankle DF/ankle equinus ⁶⁷ Decreased hip IR ⁶⁸
Patellar tendinopathy	Decreased quadriceps and hamstrings flexibility ⁶⁹ Anteriorly tilted patella ⁷⁰ Pronated foot type Supinated foot type
Patellofemoral pain syndrome	Pronated foot type ⁷¹ Hypermobile patella ⁷² Decreased quadriceps flexibility ⁷² Increased hip IR (passive and dynamic) ⁷¹ Decreased knee flexion (dynamic, e.g. jump-landing) ⁷¹ Increased knee abduction moment ^{73, 74} Increased Q angle ⁷⁵ Increased hip adduction (dynamic, e.g. running) ⁷⁶ Decreased hamstrings flexibility ⁷⁷
Iliotibial band syndrome	Increased hip adduction (dynamic, e.g. running) ⁷⁸ Increased knee IR (dynamic, e.g. running) ⁷⁸ Increased knee flexion at heel strike ⁷⁹ Increased velocity of knee IR (dynamic, e.g. running) ⁷⁹ Increased maximal foot inversion ⁷⁹
Adductor tendinopathy	Decreased total hip rotation (IR, ER) ⁸⁰

continued

Table 8.5 Cont.

Injury/condition	Associated lower limb biomechanical risk factor
Metatarsal stress fractures	Pronated foot type ¹² Increased subtalar joint inversion ⁵⁴ Supinated foot type
Tarsal stress fractures	Increased subtalar joint inversion ⁵⁴
Navicular stress fractures	Pronated foot type Ankle equinus
Fibular stress fractures	Supinated foot type Pronated foot type
Tibial stress fractures	Supinated foot type ¹²
Femoral stress fractures	Supinated foot type ¹² Decreased subtalar joint inversion ⁵⁴
Unspecified lower limb stress fractures	Pronated foot type ⁵⁴ Supinated foot type ⁵⁴ Increased hip ER ⁸¹ Leg length difference ⁸²

Evidence is included from the highest quality studies (i.e. if there is evidence from a prospective study, evidence from case-control studies are not included)

Blue = supported by prospective findings

Orange = supported by case-control or retrospective findings

Black = no/inconclusive evidence, clinical opinion

DF = dorsiflexion; PF = plantarflexion; IR = internal rotation; ER = external rotation

MANAGEMENT OF LOWER LIMB BIOMECHANICAL ABNORMALITIES

The next major theme of this chapter introduces management strategies to address biomechanical problems. There is increasing evidence about the role of therapies such as foot orthoses, footwear, taping, exercise and movement retraining to improve a patient's biomechanics. This section provides a background perspective and then chapters in Part B will address specific conditions. Here we focus mostly on interventions that affect the foot and lower leg.

Biofeedback and movement pattern retraining

In this chapter, we have described a number of suboptimal biomechanics or movement patterns which may be identified during assessment of an injured athlete. Of particular note, ideal running biomechanics and potential abnormalities have been described in some detail. In recent years, evidence supporting movement pattern retraining during running to manage lower limb injuries has begun to emerge.⁸³ This includes case series supporting visual and verbal biofeedback to reduce peak hip adduction in individuals with patellofemoral pain (PFP),^{84, 85} and to transition from rearfoot to forefoot or midfoot strike pattern in individuals with anterior chronic exertional lower leg pain.^{86, 87}

Recent evidence suggests that exercise interventions are unlikely to change actual running mechanics. This highlights the potential importance of using biofeedback to facilitate movement pattern retraining. For example, hip abductor strengthening does not appear to reduce peak hip adduction during running,⁸⁸⁻⁹⁰ a variable linked to PFP⁹¹ and iliotibial band syndrome⁹² injury risk. Importantly, movement pattern retraining can also be used for other activities such as walking, stair negotiation, squatting and sit to stand; the key is to identify a potential movement pattern which may be increasing stress on the injured tissue, and implement biofeedback to alter the movement pattern and reduce stress. This should be guided by both clinical reasoning and reported changes in symptoms from the patient. A number of suggestions for common running injuries are provided in Table 8.6. When teaching your patients movement pattern changes, it is vitally important to consider the effect this may have on other structures. For example, transition from heel to forefoot strike will reduce load on the knee and hip, but will likely increase load on the foot and ankle.⁸³

Simple clinical biofeedback methods can be divided into visual (mirror, video, etc.) and verbal (i.e. instructions). Visual biofeedback can be very important in the early stages of movement pattern retraining to ensure that the patient understands their faults and how to change them. Previously

Table 8.6 Running retraining considerations for common lower limb injuries based on current evidence and expert opinion (from Barton et al.)⁸³

Condition	Suggested retraining strategies to consider
Exertional lower leg pain	Increasing step rate, strategies to reduce overstride and land softer, and transition from rearfoot to forefoot or midfoot strike
Plantar fasciopathy	Increasing step rate, strategies to reduce overstride and land softer
Achilles tendinopathy and calf pain	Strategies to reduce overstride, increase lower limb stiffness (i.e. reduce ankle dorsiflexion) and increase hip extension
Medial tibial stress syndrome	Increasing step rate, strategies to reduce overstride and land softer, reduce hip adduction and scissoring and increase hip extension
Patellofemoral pain	Increasing step rate, strategies to reduce overstride and land softer, reduce hip adduction and internal rotation, transition from rearfoot to forefoot or midfoot strike
Iliotibial band syndrome	Reduce hip adduction and scissoring
Patellar tendinopathy	Strategies to reduce overstride and land softer, and transition from rearfoot to forefoot or midfoot strike
Hamstring injury, including proximal hamstring tendinopathy	Strategies to reduce overstride and land softer, and reduce anterior pelvic tilt and knee extension at foot strike
Gluteal tendinopathy	Increasing step rate, strategies to reduce overstride and land softer, reduce hip adduction and internal rotation, and reduce contralateral pelvic drop

published case series supporting the potential effectiveness of movement pattern retraining during running have all used some form of visual biofeedback. However, continued use in the field (e.g. using mirrors during over-ground running to reduce hip adduction) may be impractical, meaning we recommend concurrent use of verbal cues (e.g. instruction to run with knees pointing out) which can be used both in the early stages of rehabilitation and following progression into the field. Follow-up to ensure appropriate changes is essential in these circumstances, so that an athlete does not develop new maladaptive movement patterns or revert to their habitual pattern.

More complex biofeedback options are becoming easier to access in clinical practice. Clinicians can now access simple, small and affordable technologies to quantify movement analysis during assessment and provide feedback to patients during rehabilitation. One example of this is the ViMove™ (dorsaVi). This system has integrated movement and muscle sensors, and comes with software to allow the clinician to quickly analyse collected data and provide results to patients. Clinical electromyography (EMG) has been used clinically for many years to provide patients with real-time feedback regarding muscle activation patterns and magnitude. With technology in this area continuously and rapidly evolving, clinicians who are interested in incorporating more quantitative measures of biomechanics into their clinical practice should regularly review the literature and the internet for emerging technologies.

Foot orthoses

Foot orthoses (in-shoe devices) are used extensively in sports medicine to optimise lower limb function.⁹³ In this section we:

- review the various types of orthoses available
- introduce the debate about their mechanism of action
- highlight that there is high quality evidence for their effectiveness
- share three main approaches to deciding how to fit an orthosis for a specific patient.

Types of foot orthoses

Foot orthoses range in material from soft or flexible to semi-rigid devices, and may be prefabricated or custom-fabricated. Prefabricated or 'off the shelf' foot orthoses (Fig. 8.20a) are usually fabricated from materials such as EVA, polyurethane, cork or rubber. The devices' generic shape can be customised a little to the individual via heat moulding or the addition of wedges or heel raises. Prefabricated orthoses provide a quick and cheap intervention, but their limited potential for customisation and inability to achieve total plantar contact may restrict their use in some patients.

Custom-fabricated foot orthoses (Fig. 8.20b) are manufactured from a three-dimensional representation of the individual's foot using plaster impressions or laser-scanning devices. Custom foot orthoses may accommodate specific structural anomalies more effectively than prefabricated orthoses. Most often, custom foot orthoses are manufactured from polypropylene and carbon-fibre



(a)



(b)

Figure 8.20 Orthoses (a) Prefabricated or ‘off the shelf’ orthoses (b) Custom-fabricated or casted orthoses

composites. As such, they are generally more rigid than prefabricated orthoses and are often perceived to provide more effective biomechanical control. Their prescription requires a higher level of expertise and specific equipment. In the sports and exercise medicine setting, a podiatrist will typically prescribe custom-fabricated foot orthoses, which will be manufactured in a laboratory. The prescription will typically contain information regarding materials to be used, extent of accommodative postings and shoe fit. In the laboratory, a solid model of the foot is manufactured from the three-dimensional representation. Additional material is added to provide the appropriate level (degrees) of control. The material of choice is then vacuum-pressed onto the model and covered to suit the individual.

To date, no study has identified an advantage for custom-fabricated devices over prefabricated orthoses when managing lower limb conditions.⁹³ The following factors should influence the recommendation for a particular type of orthosis.

- If the patient wears a wide range of footwear during training and competition, the custom-fabricated orthosis has an advantage as it can be designed to fit a wide range of footwear; prefabricated devices may need to be modified to fit in specific footwear.
- If the patient participates in sports that involve repetitive landing, a softer or more flexible device minimises the risk of arch discomfort or blistering. Furthermore, many athletes do not tolerate large degrees of arch support from orthoses. In these circumstances, it is often helpful to reduce the arch contour.
- The difference in cost may be an important determinant for some patients, although the more expensive custom orthosis also has greater durability. It is not unusual for a custom orthosis to last 5 years or more.
- There is generally a several-week interval between fitting and supply of custom orthoses, particularly when manufactured off site. Once the patient has the

orthosis, there may be a further ‘wearing-in’ period to prevent adverse effects. This means that there may be a delay in attaining the full therapeutic benefits of the intervention. It may be appropriate to prescribe a prefabricated device for the interim period.

- Other considerations include the patient’s body mass (a more rigid orthosis may be required for people with greater body weight, due to higher forces through the foot) and personal preference.

Mechanism of action—an unfinished story

Traditional dogma was that foot orthoses controlled foot motion, and thus improved biomechanical efficiency and balanced loads on structures more appropriately. However, this is now in dispute. Some studies indicate that foot orthoses can influence foot motion,⁹⁴⁻⁹⁶ while others show that they impart no systematic effects on foot motion.^{97, 98} It is important to keep in mind that the majority of studies on foot orthoses, mechanisms of action have been conducted on healthy, asymptomatic participants, and that people with pain or injury may have different biomechanical responses to foot orthoses.

Alternative paradigms relating to shock absorption and neuromuscular effects⁹⁹ include: (i) that the cushioning effect of orthoses attenuates the impact force that occurs when the foot hits the ground;¹⁰⁰⁻¹⁰² and (ii) that the input provided by foot orthoses through their total contact with the plantar surface of the foot stimulates the neuromuscular system, which may reduce muscle activity and fatigue, and optimise performance.¹⁰³⁻¹⁰⁵ This is an exciting area of investigation which we will follow closely over the next few years.

Orthoses are effective—high quality evidence is accumulating

Irrespective of how foot orthoses exert their clinical effects, evidence from systematic reviews (level 1) and

controlled clinical trials (level 2) supports their use in a variety of conditions related to suboptimal lower limb biomechanics (Table 8.7). See Chapter 2 for a discussion of levels of evidence.

Three contemporary approaches for fitting an orthosis

Although evidence is accumulating that specific types of orthoses are effective in specific settings to treat specific clinical conditions, it is still difficult to recommend a 'gold standard' approach to prescribing orthoses. This part of our clinical approach remains an art based on limited science—as it is in many situations across healthcare. Thus, the clinician should integrate the research findings that

provide support for some approaches (see below), and also take into account the clinical assessment findings, previous clinical experience with the condition and the type of patient, as well as the patient's preferred orthosis type.

The traditional approach—Root and the goal of 'subtalar joint neutral'

Merton Root developed the functional foot orthosis in the 1950s and 1960s.¹²⁴ He proposed that subtalar joint neutral served as a standard position to evaluate structural relationships in the foot¹²⁴ and that this position represented normal foot alignment during the midstance and heel-off phases of gait.¹²⁵ Unfortunately, the alignment measures proposed by Root are unreliable and the subtalar

Table 8.7 Best available evidence for the use of foot orthoses in the management of common lower limb conditions

Condition	Type of orthosis	Level of evidence	References
<i>Prevention</i>			
Lower limb overuse conditions (stress fractures, ankle sprains, foot and ankle problems)	Prefabricated, custom-fabricated	1	Collins ⁹³
Stress fractures (femoral, tibial, unspecified)	Custom-fabricated, shock-absorbing	1	Rome ¹⁰⁶ Snyder ¹⁰⁷
<i>Treatment</i>			
Patellofemoral pain syndrome	Prefabricated	1	Barton ¹⁰⁸
Chronic musculoskeletal pain associated with pes cavus	Custom-fabricated	1	Burns ¹⁰⁹
Achilles tendinopathy	Custom-fabricated	2	Munteanu ¹¹⁰
Plantar fasciopathy	Prefabricated	2	Baldassin ¹¹¹
	Custom-fabricated	2	Baldassin ¹¹¹
			Landorf ¹¹²
			Lynch ¹¹⁴
			Martin ¹¹³
			Rome ¹¹⁵ Roos ¹¹⁶
Tibialis posterior tendinopathy	Custom-fabricated	2	Kulig ¹¹⁷
Lower limb overuse conditions (varied)	Custom-fabricated	2	Trotter ¹¹⁸
Lateral ankle sprain/chronic ankle instability	Custom-fabricated	3	Guskiewicz ¹¹⁹ Ortega ¹²⁰
Medial shin pain (medial tibial stress syndrome, shin splints)	Prefabricated	3	Louden ¹²¹
Sesamoiditis	Custom-fabricated	5	Sammarco ¹²² ACFAOM ¹²³
Peroneal tendinopathy	Custom-fabricated	5	ACFAOM ¹²³
Patellar tendinopathy	Custom-fabricated	5	ACFAOM ¹²³
Iliotibial band friction syndrome	Custom-fabricated	5	ACFAOM ¹²³

Blue = supported by systematic reviews or randomised controlled trials

Orange = supported by nonrandomised studies and case series

Black = expert opinion, clinical guidelines

Red = current evidence is contrary to common clinical consideration

joint is not in neutral during midstance.¹²⁵ Nevertheless, this approach to casting orthoses has been associated with many successful clinical outcomes and it remains one of the most common ways of prescribing orthoses.¹²⁶

Treatment direction test

Professor Bill Vicenzino¹²⁷ proposed a 'treatment direction test' (TDT) to prescribe and apply foot orthoses for lower limb musculoskeletal disorders that have a biomechanical association. The TDT complements the 'tissue-stress model' of McPoil and Hunt,¹²⁵ in that it seeks to identify symptomatic tissues under excessive loads and reduce these loads using an external physical modality. This may consist of adhesive strapping tape, temporary felt orthoses, or prefabricated foot orthoses with or without add-on wedges. The modality is selected based on what the clinician identifies as an aberrant movement pattern such as excessive or prolonged pronation.

The quality and pain-free quantity of a patient-specific aggravating activity is assessed with and without the external modality. For example, the single-leg squat is suitable for patellofemoral pain, while a single- or double-leg heel raise test is more appropriate for Achilles tendinopathy. An improvement in the quality of movement or an increase in pain-free repetitions of at least 75% indicates a high likelihood of success with subsequent prescription of orthoses. The reliability and validity of the TDT is under review.

Comfort

Nigg et al.¹⁰⁴ proposed that comfortable devices (orthoses, footwear) that support the skeleton's preferred movement path could reduce muscle activity and resulting fatigue, and thus, improve performance. Vicenzino and colleagues¹²⁸ proposed a model of prescription for prefabricated foot orthoses based on patient comfort. Once the patient reports comfort, the clinician can further modify the device to improve pain-free performance of an aggravating functional task, while aiming to maintain comfort. The first clinical trial to utilise this method reported greater than 80% success with orthoses over a 1-year period.¹²⁹ Irrespective of the long-term outcome of this branch of research, comfort should always be an important consideration for all foot orthoses prescribed. If the device is not comfortable, there is a risk of skin blistering and/or new foot pain. Also, any device that is perceived to be uncomfortable is likely to lead to poor patient compliance.

Footwear as a therapy rather than as a cause of injury!

When considering any intervention addressing suboptimal lower limb biomechanics, the potential influence of footwear should be considered (see also 'Clinical assessment

of footwear' earlier in this chapter). Consider how the current footwear influences the patient's condition, how footwear interacts with other treatment (e.g. foot orthoses) and whether altering footwear characteristics can help treat the condition.

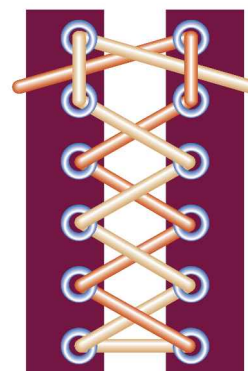
Footwear fit is particularly important when prescribing orthoses. If an orthosis is added to a shoe that has inadequate room, it may result in forefoot pain or limit the time the orthosis is worn. Remember that athletes often wear a range of shoes in training and competition—check all of them!

If the patient's shoe has suboptimal motion control (e.g. minimal heel counter stiffness) leading to excessive foot pronation, shoe replacement may be the best treatment. Footwear with greater heel counter stiffness, midfoot sole rigidity, adequate lacing and possibly a multiple density sole (i.e. increased density on the medial aspect of the shoe) may reduce excessive foot motion. Footwear support can also be enhanced by education on how to complete loop lacing (Fig. 8.21). Such changes in relation to footwear may reduce the need for foot orthoses.

Taping

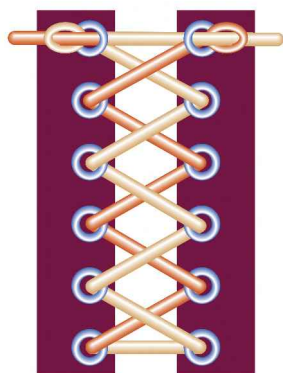
Adhesive strapping tape (Chapter 17) is a temporary intervention to address lower limb biomechanical issues. Anti-pronation tape is commonly used to treat plantar fasciopathy (Chapter 43) and patellofemoral pain (Chapter 36).

One of the most commonly used techniques for lower limb musculoskeletal pain is the Augmented low-Dye. This consists of the original low-Dye¹³⁰ (anchor running horizontally around the heel from the head of the fifth to the head of the first metatarsal, with strips of tape under



(a)

Figure 8.21 (a) Lock-lacing to prevent heel slippage. Lace the shoe normally until the second set of eyelets. Then feed the laces into the top eyelet on the same side. Now cross each lace over, and feed through the loop formed between the first and second eyelet on the opposite side. Laces should then be pulled and tied as normal



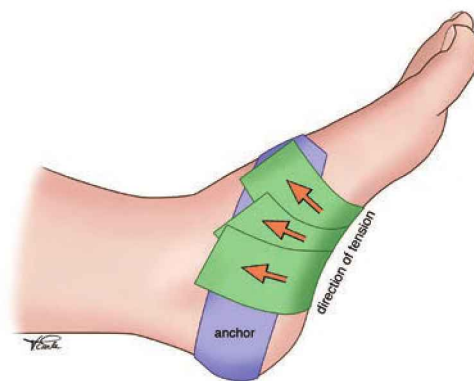
(b)

Figure 8.21 (cont.) (b) Loop to create a snug fit. After lacing, put each lace end back through the last hole to create a small loop on the top side of the shoe. Then thread each loose end through the loop on the opposite side and tighten

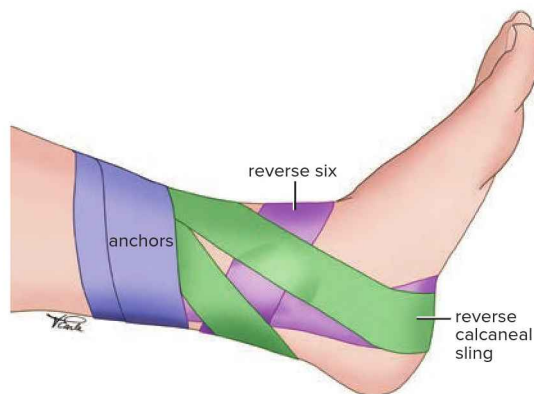
the foot directed laterally to medially to support the arch, Fig. 8.22a), which is augmented with reverse sixes and calcaneal slings anchored to the lower leg¹²⁷ (Fig. 8.22b).

Anti-pronation tape has both biomechanical and neuromuscular effects during static and dynamic tasks.¹³¹ Specifically, it increases navicular and medial longitudinal arch height, reduces tibial internal rotation and calcaneal eversion, alters patterns of plantar pressure, and reduces activity of particular leg muscles.¹³¹

Techniques such as patellar taping may also alleviate symptoms associated with PFJ biomechanics.^{132, 133} It is important to be aware of skin breakdown associated with prolonged use of tape, particularly in athletes involved in vigorous activity and implement appropriate strategies to prevent this.



(a)



(b)

Figure 8.22 (a) Low-Dye taping can be used to restrict subtalar pronation (b) The addition of calcaneal slings and reverse sixes anchored to distal leg can augment low-Dye taping¹²⁷

A team approach to biomechanical correction

In this box, we emphasise that clinicians working together can address biomechanical factors more effectively than one clinician working alone.

Exercises and functional retraining

Exercises and functional retraining should always be considered when managing lower limb biomechanical issues. All clinicians in sport and exercise medicine should appreciate the influence of muscle imbalance to biomechanical abnormalities. Thus, excessive tightness of muscles such as the psoas, tensor fascia latae, hip adductors, hamstrings and gastrocnemius can be addressed using exercises such as static and dynamic stretches, in conjunction with other modalities such as massage, heat or dry needling.

Muscle weakness or incoordination requires strengthening and retraining exercises, as follows.

- Dysfunction of the abdominals, gluteus medius and minimus, hip external rotators, vastus medialis obliquus and tibialis posterior should be considered in the patient with suboptimal lower limb biomechanics. While non weight-bearing exercises may be used initially, they should be progressed to functional weight-bearing positions as soon as possible.
- Once optimal static muscle activation has been achieved, exercises can be progressed by adding lower limb movements, resistance (e.g. dumbbells, resistance bands) or stability challenges (e.g. exercise balls, single-limb stance).

continued

- Motor control exercises of the hip and foot are useful to promote optimal alignment of the lower limb in the sagittal, frontal and transverse planes, as well as ideal muscle recruitment patterns.
- Exercises that incorporate the lower limb in its closed kinetic chain function include the single-leg squat, single-leg heel raise, single-leg stance and arch form.
- The next stage involves integration of this new control into functional activities, such as running, landing or sport-specific skills. Small components of the overall movement should be incorporated initially, along with specific instruction and feedback. These movements may need to be performed slowly to allow them to be integrated successfully into the functional activity (Chapter 18).

Attend to the entire kinetic chain

It is important to consider the potential causes and effects of suboptimal biomechanics on the entire neuromusculoskeletal system. For example:

- ankle joint stiffness may contribute to altered biomechanics during gait or occur as a consequence of calf muscle imbalance
- joint pain or stiffness can be addressed with active or passive mobilisation of peripheral or vertebral joints or 'mobilisation with movement'¹³⁴
- increased neural mechanosensitivity can be managed with appropriate exercises, as well as correction of possible causes such as spinal hypomobility.

Other interventions aimed at reducing pain, such as pharmacotherapy, electrotherapy and dry needling may facilitate optimal performance of exercises (Chapter 17).

UPPER LIMB BIOMECHANICS

with W. BEN KIBLER

Correct biomechanics is as important in upper limb activities as it is in lower limb activities. For example, repeated throwing places tremendous stresses on the upper limb, especially the shoulder and elbow joints.¹³⁵ Throwing, however, is a 'whole body activity', involving the transfer of momentum from the body to the object being thrown. The following section uses baseball pitching as a model; however, the principles described are relevant to other overhead sports activities, such as the javelin throw, volleyball spike and tennis serve.

The biomechanics of throwing

Throwing is a whole body activity that commences with drive from the large leg muscles and rotation of the hips, and progresses through segmental rotation of the trunk and shoulder girdle. It continues with a 'whip-like' transfer of momentum through elbow extension and through the small muscles of the forearm and hand, transferring propulsive force to the ball.

The skilled clinician should assess both the scapulohumeral and the truncal mechanics in a throwing athlete. The role of the scapula in throwing is discussed in more detail below, and the back, trunk and hips serve as a centre of rotation and a transfer link from the legs to the shoulder.

Throwing can be divided into four phases:

- | | | |
|--------------------------------|---|-------------------|
| 1. Preparation/wind-up | } | 80% time sequence |
| 2. Cocking | | |
| 3. Acceleration | | 2% time sequence |
| 4. Deceleration/follow-through | | 18% time sequence |

Wind-up

Wind-up (Fig. 8.23) establishes the rhythm of the throw. During wind-up, the body rotates so that the hip and shoulders are at 90° to the target. The major forces arise in the lower half of the body and develop a forward-moving 'controlled fall.' In pitching, hip flexion of the lead leg raises the centre of gravity. The wind-up phase lasts 500–1000 milliseconds. During this phase, muscles of the shoulder are relatively inactive.

Cocking

The cocking movement (Fig. 8.24) positions the body to enable all body segments to contribute to ball propulsion. In cocking, the shoulder moves into abduction through full horizontal extension and then into maximal external rotation. When the scapula is maximally retracted, the acromion starts to elevate. With maximal external rotation, the shoulder is 'loaded', with the anterior capsule coiled tightly in the apprehension position storing elastic energy. The internal rotators are stretched.¹³⁶ At this stage, anterior joint forces are maximal and can exceed 350 newtons (N).

Toward the end of cocking, the static anterior restraints (anterior inferior glenohumeral ligament and anterior inferior capsule) are under the greatest strain. Because of the repetitive nature of throwing, these structures can become attenuated, leading to subtle instability.¹³⁷ In the trunk, tensile forces increase in the abdomen, hip extensors and spine, with the lead hip internally rotating just prior to ground contact.

The cocking phase ends with the planting of the lead leg, with the body positioned for energy transfer through the legs, trunk and arms to the ball. This phase also lasts 500–1000 milliseconds. The wind-up and cocking phases



Figure 8.23 Throwing—wind-up

together constitute 80% of the duration of the pitch (approximately 1500 milliseconds).

Shoulder cocking continues with the counterclockwise rotation of the pelvis and trunk (when a right-handed thrower is viewed from above), which abruptly places the arm behind the body in an externally rotated position.

Lateral trunk flexion determines the degree of arm abduction. When viewed in the frontal plane, the relative abduction of the humerus to the long axis of the trunk is a fairly constant 90–100°, regardless of style. The overhand athlete leans contralaterally, while the ‘side-arm’ or ‘submarine’ thrower actually leans toward the throwing arm. Rotation of the trunk also aids in abduction. Although the muscles of the shoulder produce little abduction during the early cocking phase of a well-executed throw, the periscapular muscles are quite active. The force couple between the upper trapezius and serratus initiates acromial elevation and the lower trapezius maintains elevation at abduction angles greater than 65°.

Acceleration

The acceleration phase (Fig. 8.25) is extremely explosive. It consists of the rapid release of two forces—the stored elastic force of the tightly bound fibrous tissue of the



Figure 8.24 Throwing—cocking

capsule, and forceful internal rotation from the internal rotators (subscapularis, pectoralis major, latissimus dorsi and teres major). This generates high forces at the glenohumeral articulation,¹³⁸ creating high demands on the cuff musculature to keep the humeral head located in the glenoid.

Large muscles outside the rotator cuff are responsible for the subsequent acceleration of the arm. This includes muscles of the anterior chest wall as well as the muscles and fascia that surround the spine. The critical role of the muscles controlling scapulothoracic motion—scapular positioning and stabilisation against the thorax—is discussed below.

At the shoulder, acceleration is the shortest phase of the throwing motion, lasting only 50 milliseconds (2% of the overall time). In both the acceleration and the late cocking phases, muscle fatigue (which is accelerated if there is mild instability due to attenuated static restraints) can lead to loss of coordinated rotator cuff motion and, thus, decreased anterior shoulder wall support.

The acceleration phase concludes with ball release, which occurs at approximately ear level. The movements involved in acceleration place enormous valgus forces on the elbow, which tends to lag behind the inwardly rotating shoulder.



Figure 8.25 Throwing—acceleration

Deceleration/follow-through

Not all the momentum of the throw is transferred to the ball. In the deceleration/follow-through phase (Fig. 8.26), large forces pull forward on the glenohumeral joint following ball release, which stresses the posterior shoulder structures. During this time both intrinsic and extrinsic shoulder muscles fire at significant percentages of their maximum, attempting to develop in excess of 500 N to slow the arm down. The force tending to pull the humerus out of the shoulder socket can exceed 500 N (roughly equivalent to 135 kg). The eccentric contraction of the rotator cuff external rotators decelerates the rapid internal rotation of the shoulder, as does eccentric contraction of the scapular stabilisers and posterior deltoid. In the properly thrown pitch, the spine and its associated musculature have a significant role as a force attenuator.

Toward the end of the pitching motion, the torso, having decelerated so the arm could acquire kinetic energy in the arm acceleration phase, begins to rotate forward. The forward rotation of this larger link segment helps to reacquire some of this energy. This theoretically reduces the burden on the serratus anterior and other stabilisers, which are attempting to eccentrically maintain



Figure 8.26 Throwing—deceleration/follow-through

the position of the scapula and maintain the humeral head within the glenoid.

In addition to the high stresses on the posterior shoulder structures, this phase places large stresses on the elbow flexors that act to limit rapid elbow extension. This phase lasts approximately 350 milliseconds and constitutes approximately 18% of the total time.

The role of the trunk in throwing is clear. When trunk motion is inhibited or the potential GRF reduced, throwing velocity is markedly lower. With a normal overhead throw rated at 100%, peak velocities dropped to 84% when a forward stride was not allowed and down to 63.5% and 53.1% when the lower body and lower body plus trunk were restricted, respectively.¹³⁹ Peak ball-release velocities attained by water polo players are approximately half the velocity that a thrown baseball might reach on land where a GRF can be generated.

The kinetic chain

The 'kinetic chain' is a term clinicians use to make the point that optimal function of a distal structure relies on well-functioning proximal elements. Consider the hand/fingers in throwing as a 'distal' structure. The kinetic chain that underpins successful throwing includes the

lower limb (the very beginning of the 'chain'), the muscles that envelop the torso, the scapula and shoulder muscles and finally the upper arm and forearm that connect to the hand (Fig. 8.27). It does not require much imagination to see these elements (links) working like a 'whip' or a dynamic 'chain' to allow very speedy release of a baseball, cricket ball or tennis ball (with the racquet extending the chain even further).

Let us illustrate the concept more specifically using the scapula as a model. It links the proximal-to-distal sequencing of velocity, energy and forces that optimise shoulder function. For most shoulder activities this sequencing starts at the ground. Individual body segments, or links, move in a coordinated way to generate, summate and transfer force through various body segments to the terminal link. Large proximal body segments provide the bulk of the force.

Why is the kinetic chain concept important for clinicians? Because it highlights that treatment cannot merely address local symptoms. Right shoulder tendon pain in a right arm bowling/throwing athlete may arise because of calf muscle weakness, hip joint tightness and in coordination of the scapular movements. Clinicians must look for causes of symptoms along the entire kinetic chain and treat all contributing links.

Normal biomechanics of the scapula in throwing

In recent years, the importance of the scapula in normal throwing biomechanics has been increasingly recognised. For optimal shoulder function and to decrease injury risk, the scapula must move in a coordinated way (Fig. 8.28). This section outlines Ben Kibler's¹⁴⁰ description of the role of the scapula in throwing (Table 8.8). If the clinician understands the normal scapular biomechanics,

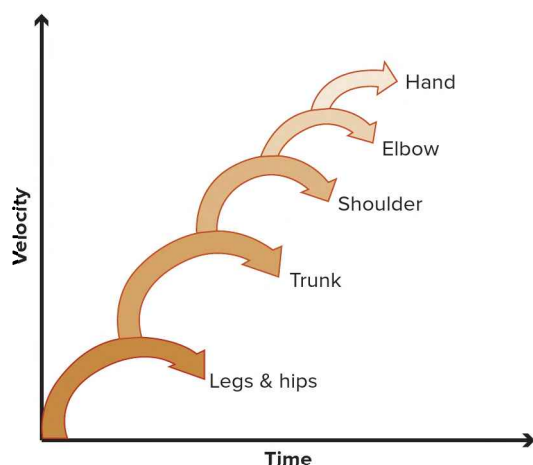


Figure 8.27 The kinetic chain in throwing

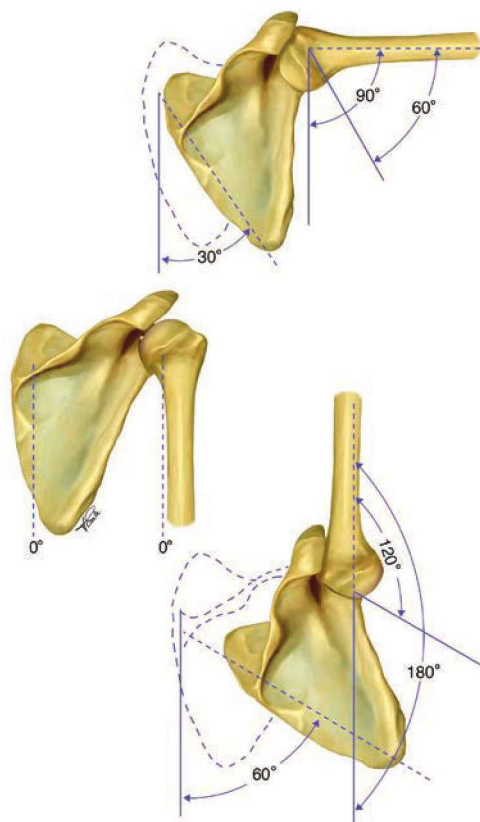


Figure 8.28 Normal scapulothoracic rhythm allows the scapula to rotate upwardly during abduction, bringing the glenoid fossa directly under the humeral head to lend stability to the glenohumeral joint

Table 8.8 Scapular function in normal shoulder mechanics

1. Provides a stable socket for the humerus
2. Retracts and protracts along the thoracic wall
3. Rotates to elevate the acromion
4. Provides a base for muscle attachment
5. Provides a key link in the kinetic chain

he or she will be then able to detect abnormal scapular biomechanics in patients with upper limb injuries.

The scapula provides a stable socket for the humerus

In normal shoulder function, the scapula forms a stable base for glenohumeral articulation. The glenoid is the socket of the ball-and-socket glenohumeral joint. Thus, the scapula must rotate as the humerus moves so that the centre of rotation of the glenohumeral joint remains optimal throughout the throwing or serving motion.

This coordinated movement keeps the angle between the glenoid and the humerus within the physiologically tolerable or 'safe zone', which extends about 30° of extension or flexion from neutral. In this range, there is maximal 'concavity/compression' of the glenohumeral joint, and the muscle constraints around the shoulder are also enhanced. The maximal concavity/compression results from the slightly negative intra-articular pressure of the normal joint, with optimal positioning of the glenoid in relation to the humerus and coordinated muscle activity.

The scapula must retract and protract along the thoracic wall

In the cocking phase of throwing (as well as in the tennis serve and swimming recovery), the scapula retracts (see above). Once acceleration begins, the scapula protracts smoothly laterally and then anteriorly around the thoracic wall to keep the scapula in a normal position relative to the humerus and also to dissipate some of the deceleration forces that occur in follow-through.

The scapula rotates to elevate the acromion

As almost all throwing and serving activities occur with a humerus-to-spine angle of between 85° and 100° of abduction, the scapula must tilt upwards to clear the acromion from the rotator cuff.

The scapula provides a base for muscle attachment

Stabilising muscles attach to the medial, superior and inferior borders of the scapula to control its position and motion. The extrinsic muscles (deltoid, biceps and triceps) attach along the lateral aspect of the scapula and perform gross motor activities of the glenohumeral joint. The intrinsic muscles of the rotator cuff attach along the entire surface of the scapula and work most efficiently with the arm between 70° and 100° of abduction. In this position, they form a 'compressor cuff' locating the humeral head into the socket.

The scapula provides a key link in the kinetic chain

The scapula is pivotal in transferring the large forces and high energy from the legs, back and trunk to the arm and the hand. Forces generated in the proximal segments are transferred efficiently and are regulated as they go through the funnel of the shoulder when the scapula provides a stable and controlled platform. The entire arm rotates as a unit around the stable base of the glenohumeral socket.

Thus, the scapula performs various interrelated functions to maintain the normal glenohumeral path and

provide a stable base for muscular function. Abnormalities in scapular function that predispose to injury are discussed below.

Abnormal scapular biomechanics and physiology

The scapular roles can be altered by many anatomical factors to create abnormal biomechanics and physiology, both locally and in the kinetic chain (Table 8.9).

Clinical significance of scapular biomechanics in shoulder injuries

Abnormal shoulder biomechanics can compromise normal shoulder function. This observation has been given various descriptive titles, such as 'scapulothoracic dyskinesis', 'floating scapula' or 'lateral scapular slide'. It is important for the clinician to recognise that these are merely titles for the same phenomenon, that is, abnormal scapular function. We provide examples of how abnormal biomechanics can cause shoulder and elbow problems.

Lack of full retraction of the scapula on the thorax destabilises the cocking point and prevents acceleration out of a fully cocked position. Lack of full scapular protraction increases the deceleration forces on the shoulder and alters the normal safe zone between the glenoid and the humerus as the arm moves through the acceleration phase. Too much protraction because of tightness in the glenohumeral capsule causes impingement as the scapula rotates down and forward. These cumulatively lead to abnormalities in concavity/compression due to the changes in the safe zone of the glenohumeral angle.

Loss of coordinated retraction/protraction in throwing opens up the front of the glenohumeral joint and, thus, provides an insufficient anterior bony buttress to anterior translation of the humeral head. This increases shear stress on the rest of the anterior stabilising structure—the labrum and glenohumeral ligaments—which further decreases the stability of the glenoid for the rotating humerus.

Lack of acromial elevation leads to impingement in the cocking and follow-through phases. Impingement can also occur secondary to painful shoulder conditions that inhibit the function of the serratus and lower trapezius muscles. As these muscles normally act as a force couple to elevate the acromion, their inhibition commonly causes impingement. Thus, detecting and, if necessary, reversing serratus and trapezius inhibition is an important step in treating shoulder conditions.

If the scapula is unstable, the lack of an anchor affects the function of all scapula muscles. Muscles without a stable origin cannot develop appropriate or maximal torque and are predisposed to suffering muscular imbalance. If the scapula is truly unstable on the thoracic

Table 8.9 Alterations to scapular function

Scapular function alteration	Effect on scapular function
<i>Anatomical factors</i>	
Cervical spine lordosis	Excessive scapular protraction—leads to impingement with elevation
Thoracic spine kyphosis	Excessive scapular protraction—leads to impingement with elevation
Shoulder asymmetry (i.e. drooping of the shoulder or 'tennis shoulder')	Impingement/muscle function and fatigue
Injuries of scapula, clavicle	Alters orientation of scapula, length of clavicular strut—painful conditions that inhibit muscle function
<i>Abnormalities in muscle function</i>	
Overuse, direct trauma, glenohumeral causes (instability, labral lesions, arthrosis)	Muscle weakness or force couple imbalances. Serratus anterior and lower trapezius particularly susceptible. Can be a non-specific response to a variety of glenohumeral pathologies (this can be seen as analogous to the knee in that weakness of the vastus medialis obliquus can result in patellofemoral syndrome)
Glenohumeral inflexibility, posterior (capsular or muscular)	Limits smooth glenohumeral joint motion and creates wind-up effect so that the glenoid and scapula get pulled forward and inferiorly by the moving arm. This leads to excessive protraction, which, in turn, holds the scapula and, importantly, the acromion inferiorly and, thus, makes it prone to impingement
Nerve injury (causes less than 5% of abnormal muscle function in shoulder problems)	Long thoracic nerve—serratus anterior function inhibited. Accessory nerve—trapezius function inhibited

wall, as in spinal accessory nerve palsies or in extremely inhibited muscles, then the muscle origins and insertions are effectively reversed, and the distal end of the muscle becomes the origin. The scapula is then pulled laterally by the muscle, which contracts from the more stable distal humeral end rather than from the proximal scapular end. A further problem of the unstable scapula is that it does not provide a stable base for glenohumeral rotation during link sequencing. Therefore, the arm works on an unstable platform and loses mechanical efficiency.

One of the most important scapular biomechanical abnormalities is the loss of the link function in the kinetic chain. The kinetic chain permits efficient transfer of energy and force to the hand. The scapula and shoulder funnel forces from the large segments, the legs and trunk, to the smaller, rapidly moving small segments of the arm.

Scapular dysfunction impairs force transmission from the lower to the upper extremity. This reduces the force delivered to the hand or creates a situation of 'catch up' in which the more distal links have to overwork to compensate for the loss of the proximally generated force.

The distal links have neither the size, the muscle cross-sectional area, nor the time in which to develop these larger forces efficiently. For example, a 20% decrease in kinetic energy delivered from the hip and trunk to the arm necessitates an 80% increase in muscle mass or a 34% increase in rotational velocity at the shoulder to deliver the same amount of resultant force to the hand. Such an adaptation would predispose to overload problems.

This explains why injuries apparently unrelated to the upper limb, for example, decreased push off due to Achilles tendinopathy, decreased quadriceps drive after a muscle strain, or decreased segmental trunk rotation secondary to thoracic segmental hypomobility, can affect upper limb throwing mechanics and predispose to further or more serious, upper limb injury.

Changes in throwing arm with repeated throwing

Repeated throwing causes adaptive changes to gradually develop in the shoulder and elbow. Changes occur in flexibility, soft tissue/muscle strength and bony contour.

At the shoulder, long-term throwing athletes have increased range of external rotation. This arises because of the repeated stress to the anterior capsule in the cocking phase, and stretch or breakdown in the anterior static stabilisers of the shoulder joint (the inferior glenohumeral ligaments). This may compromise the dynamic balance that exists between shoulder function and stability. The combination of increased shoulder external rotation range of motion and breakdown of the static stabilisers may lead to anterior instability of the shoulder and secondary impingement.

The normal strength ratio of internal rotators to external rotators is approximately 3:2, but in throwers this imbalance is exaggerated and, over time, lack of external rotation strength may increase vulnerability to injury. These dynamic changes in the shoulder joint highlight the need for a structured exercise program to prevent or correct muscle imbalances.

Throwing also produces structural changes at the elbow. Due to the valgus stress applied in the throwing action, there is a breakdown of the medial stabilising structures (medial collateral ligament, joint capsule, flexor muscles). This leads to the development of an increased carrying angle at the elbow.

Less frequently, the eccentric overload on elbow structures causes anterior capsular strains, posterior impingement or forearm flexor strains and subsequently, a fixed flexion deformity.

Common biomechanical abnormalities specific to throwing

One of the most common biomechanical problems seen in throwing sports such as cricket, baseball and water polo is caused by the throwing athlete 'opening up too soon'. Normally the body rotates out of the cocking phase when the arm is fully cocked (externally rotated). If the body opens up too soon, the arm lags behind and is not fully externally rotated. This results in increased stress to the anterior shoulder structures and an increased eccentric load to the shoulder external rotators. It also results in increased valgus stress at the elbow.

The other common abnormality seen in throwing athletes is known as 'hanging', which is a characteristic sign of fatigue. Decreased shoulder abduction leads to dropping of the elbow and a reduction in velocity. There is an associated increase in the likelihood of injury, particularly to the rotator cuff as well as to the shoulder joint and the elbow. It is normally related to excessive intensity, frequency or duration of activity.

In baseball, the type of pitch is determined by the spin imparted onto the ball by the hands and fingers at ball release. The normal follow-through involves forearm pronation. In 'breaking' pitches, the forearm is relatively supinated at release and then pronates. 'Breaking' pitches are associated with an increased risk of injury. Some pitchers incorrectly forcefully supinate against the normal pronation of follow-through.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e